SAR Image Classification Using Greedy Hierarchical Learning With Unsupervised Stacked CAEs

Zhensheng Sun, Juan Li, Peng Liu, Weijia Cao, Tao Yu, and Xingfa Gu

***Abstract*— Synthetic aperture radar (SAR) can provide stable data source for earth observation due to its advantages of all day and night, all-weather, and strong penetration. SAR image classification as a fundamental procedure has been proved its great value in plenty of remote sensing applications. Conventional classification algorithms mainly rely on hand-designed features, which are susceptible to widespread coherent speckle noise and geometric distortion in high-resolution SAR images. Inspired by the recent impressive success in data mining and deep learning, a greedy hierarchical convolutional neural network (GHCNN) is developed. It aims at obtaining optimized feature representation, relieving the effect of speckle noise, and promoting the local pattern recognition of geometric distortion in single-polarized SAR image classification. First, a series of convolutional autoencoders (CAEs) is trained in the greedy layer-wise unsupervised strategy. This step provides an unbiased regularizer and *a priori* distribution derived from large volumes of unlabeled SAR patches. Then, to optimize multiple parameter subspaces globally, several CAEs are coupled together to form a deeper hierarchical structure in a stacked and unsupervised fashion. Afterward, a convolutional network with identical topology inherits the pretrained weights. After supervised finetuning, it realizes class prediction. Synchronously, t-distributed stochastic neighbor embedding (t-SNE) algorithm is applied to monitor the efficiency of feature representation during the training period. Experimental results demonstrate that the proposed method has competitive advantages over involved contrast methods.**

***Index Terms*— Convolutional neural network (CNN), hierarchical parameter-oriented learning, image classification, local pattern recognition, stacked convolutional autoencoders (SCAEs), synthetic aperture radar (SAR).**

I. Introduction

As ONE of the most important active remote sensing technologies, synthetic aperture radar (SAR) provides a new perspective for earth observation with unique electromagnetic spectrum and imaging mechanism. Due to its advantages of all day and night, all-weather, and strong penetration capability, the temporal and spatial dimensions of remote sensing data source have been greatly expanded. SAR image classification as a fundamental procedure for plenty of applications has been proved its great value in tsunami damage assessment [1], sea ice detection [2]–[4], oil spill pollution [5], [6], terrain surface classification [7]–[9], and agricultural monitoring [10], [11].

The workflow of remote sensing image classification usually follows: image preprocessing, feature extraction, classifier designing, result postprocessing, and accuracy evaluation. Owing to the unique imaging mechanism and coherent speckle noise, one of the biggest challenges for SAR image classification is informative and distinguishing feature extraction. The difference of backscattering characteristic of SAR images is mainly reflected in gray scale and texture. However, gray scale in single pixel is not able to capture abundant structural information, especially in high-resolution SAR images. Therefore, in order to characterize spatial context information for SAR images, feature extraction methods focus on texture-based local pattern recognition, including statistical analysis [12]–[14], random field models [15]–[18], mathematical morphology [19], [20], and signal transformations [21], [22].

The commonly used statistical descriptor, called gray-level co-occurrence matrix (GLCM) [12], estimates image properties from a distribution matrix in several defined offsets and directions. But GLCM algorithm usually requires gray-level compression to reduce the time complexity. This operation causes severe SAR image degradation and impedes the maintenance of detailed information. To partially solve the computationally time-consuming problem, a designed multilevel local pattern histogram (MLPH) descriptor [14] was developed. In this method, the intensity distributions and homogenous local patterns are described in a multiscale sliding window. In the perspective of stochastic process, the texture of SAR images can be regarded as a display of 2-D random field. In other words, the texture can be quantitatively described by random field models, such as Markov random field (MRF) [15]–[17]. MRF reveals texture aggregation relationships among the pixels in a neighborhood, which takes into account various orientations and offsets. The complexity and convergence of model parameters are the main bottlenecks of this kind of methods. Mathematical morphology considers that the texture of SAR images can be described as the type, quantity, and repeatability of elements. Morphological

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skeleton [19], [20] ensures regional and global representation of the precise shapes, which provides simplification and crucial information. Compared with the spatial domain, the frequency domain processing for SAR images offers unique advantages. Wavelet transformation [22] is an efficient way to obtain a multiresolution representation of SAR images. It is consistent with human visual perception and combines space-frequency domain to analyze texture features. In conclusion, the aforementioned methods, to some extent, suffer tedious parameter adjustment and poor generalization ability for SAR image classification, which set limitations with the sharp increase of image details. Furthermore, some abovementioned methods demand a lot of prior knowledge as well. The complex mixing of various land cover types has also not been well solved, especially in SAR images with large coverage.

In recent years, deep learning has achieved a breakthrough in a wide range of fields with its self-learning, robustness, and generalization capacities [23]–[27]. Deep learning is a feature learning method based on the hierarchical representation and advanced abstraction of deep neural networks (DNNs), especially convolutional neural network (CNN). It is far superior to those of hand-crafted feature extractors [23]. Specifically, a novel technique for SAR image classification in urban areas was developed in [28], which used a stacked autoencoder (SAE) network to transform the data set into a compact representation. In addition, another approach [29] was based on a deep convolutional autoencoder (DCAE), where a classification framework was composed of eight layers to optimize features and classify. Meanwhile, a two-stream feature extraction algorithm using CNNs to perform multifeature fusion was proposed in [30] and [31]. Wang *et al.* [30] designed separated convolutional streams to combine intensity and gradient amplitude features of SAR images. In another method, Geng *et al.* [32] proposed a deep supervised and contractive neural network (DSCNN) for SAR image classification. A multiscale model which consists of several different feature descriptors such as Gabor and HOG was built to produce efficient feature representation. Moreover, a patch-sorted DNN (PSDNN) method [33] was proposed for SAR image classification, which applied a CNN combined with a dual-sparse autoencoder (DSAE) to obtain high-level spatial and structural features.

However, most of the deep model transfer applications rarely take into account the adaptive optimization for the intrinsic differences among multisource remote sensing images. The coherent imaging mechanism of SAR data introduces many distinctions to the image sampling units. Compared with optical remote sensing or natural images, the characteristics of SAR images are mainly shown in the aspects of spectrum, texture, noise, distortion, ambiguity, and volume. For instance, the lack of spectrum is a big limitation for SAR images. It is a major restriction for the classification of different land cover types. The dominant discrimination of single-polarized SAR images exists in the intensity of backscattering and texture. But SAR image quality is with some degrees of uncertainty in intensity and texture. The image quality is greatly influenced by the observation angle, platform orbit, target structure, and material. Furthermore, the data volume of satellite SAR is limited in most cases. Considering these differences in quantity and quality, there are some difficulties that stand in the way for SAR image classification, including the lack of optimized features and limited data volume. For image pattern recognition, most of the state-of-the-art deep models originate from computer vision. Transfer learning mechanism makes it much easier for optical images to realize the inheritance of advanced abstract features. As for SAR data, there are still some limitations to be considered and resolved. On the one hand, the direct transfer application of traditional deep models is not suitable for single-polarized SAR images. The spectrum-oriented computational topology and kernels will be challenged by the spectral limitation and neighboring ambiguity. On the other hand, the training of deep models thirsts for a big data set with clean labels [35]. But the supervised process for a big SAR data set with clean label is extremely difficult and expensive. The limited volume of SAR data is not enough to support the fully supervised large-scale training. In addition, many previous researches use the narrow and ideal SAR images to test the algorithms. In fact, it is inefficient for many deep models to process the broad and complex SAR images. The traditional single-stream deep models are often faced with many practical limitations. For the intelligent remote sensing in the future, such as the intelligent on-board processing system for SAR satellite or unmanned aerial vehicle (UAV), the design of lightweight optimized models is highly necessary. In most remote sensing applications, excluding the negative impact of limited data sets will bring practical significance.

Meanwhile, the training quality of deep models in remote sensing image classification is an implicit concern, which is always ignored by previous work. Although several researches above have achieved impressive results, in fact, there still remain many uncertainties in the empirical parameter tuning. For example, overfitting and local minima [34] are the common training problems in the highly nonconvex situation. And besides, these potential challenges always have negative impacts on the convergent velocity of intricate parameter space. In most cases, the processes of empirical parameter adjustment are extremely inefficient, which greatly reduces the practical value of many deep learning algorithms. For the past few years, the unsupervised pretraining strategy has been proposed in many deep models such as deep belief network (DBN) and SAE. This strategy introduces great value in model adaptation and optimization, which indicates a better yielding to global minima [36], [37]. Many state-of-the-art artificial networks obtain huge improvements, but it has not been widely used in the convolutional networks. It has great potential to partially alleviate some of the restrictions and problems suffered in the complex image classification.

Based on an overall consideration, we propose a greedy hierarchical CNN (GHCNN) model. It aims to solve the existing issues of SAR image classification and realize an efficient combination of deep learning algorithms. Three original contributions are reflected in this article.

1) A GHCNN model is proposed to realize an efficient patch-based classification for single-polarized SAR images. Compared with the classical single-stream CNN

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model, the GHCNN develops a multistream topology to integrate microscale contextual information in higher performance. Multilevel convolution and pooling/unpooling operators build a hierarchical filtering pattern, which greatly relieves the effect of speckle noise. And at the same time, the 2-D spatial locality is well preserved. As for the broad SAR images, the lightweight and multistream topology also makes the proposed model more competitive in processing efficiency.

2) A greedy hierarchical learning strategy is developed to enhance the feature representation for single-polarized SAR images. Several CAEs are stacked and trained using randomly sampled SAR patches in a greedy layer-wise approach. These CAEs capture pervasive structural SAR features and produce intermediate representation of the input distribution pattern automatically. This greedy learning strategy provides multilevel CAE kernels to realize an optimized SAR image pattern perception, including backscattering intensity, texture, speckle noise, and distortion. It dramatically reduces the empirical nonorientation parameter training from scratch and guides the parameter space toward the tendency of minimal basins. Therefore, the whole model greatly promotes the discrimination of feature representation.
3) A semisupervised training mechanism is adopted to provide efficient use of limited labeled SAR patches on large-scale model. To balance the asymmetry between limited SAR data volume and deep model training, the parameter-oriented stage follows an unsupervised approach. And the classified stage introduces supervised finetuning using labeled SAR patches derived from Regions of Interest (ROIs). This mechanism takes full advantage of much unlabeled SAR data in a low-cost approach. In addition, using this augmented training mechanism can also improve the model robustness and generalization.

The remainder of this article is organized as follows. In Section II, the core idea and overall architecture of GHCNN pretrained with SCAEs are described in detail. Experimental results and maturity assessment are discussed in Section III. Section IV draws the conclusions and proposes the future work.

II. Greedy Hierarchical Learning With Convolutional Autoencoders (CAEs)

A. Overall Pipeline for SAR Image Classification

In this section, the whole pipeline proposed for SAR image classification is presented. First of all, SAR patches randomly sampled without labels are collected to train CAEs in a greedy hierarchical fashion. A series of unsupervised unit CAEs stacked together makes up a deep architecture, which sets an optimal stage for the weights of convolutional hierarchy. Afterward, the convolutional layers with identical topology as SCAEs are initialized with the pretrained weights. In the end, supervised fine-tuning is applied to the GHCNN, which involves with feature extraction and final classification. The whole framework is depicted in Fig. 1, including three pivotal backbones: greedy layer-wise CAEs, stacked CAEs (SCAEs), and GHCNN initialization and supervised finetuning.

B. Greedy Layer-Wise CAEs

Autoencoder is an unsupervised learning algorithm, which aims to reproduce the input signal as similar as possible, so that the output signal yields to the input [38]. It relies on end-to-end encoder and decoder training to achieve the approximation. The AE model aims to learn an approximate function computed as

$$h_{\boldsymbol{k},\boldsymbol{b}}(\boldsymbol{x}) \cong \boldsymbol{x} \tag{1}$$

where x is the input signal, $h(\cdot)$ denotes the autoencoders function, k and b stand for the weights and biases between layers, respectively. For the ith sample $x_i \in T^d$, the loss function is defined as

$$L\left(\boldsymbol{k},\boldsymbol{b};\boldsymbol{x}_i,h_{\boldsymbol{k},\boldsymbol{b}}(\boldsymbol{x}_i)\right)=\frac{1}{2}\left\|h_{\boldsymbol{k},\boldsymbol{b}}(\boldsymbol{x}_i)-\boldsymbol{x}_i\right\|^2 \tag{2}$$

where $L(\cdot)$ is the loss function for a single sample. It subsequently resorts to optimization algorithms for an appropriate minimization over the training space T^d. And $h_{\boldsymbol{k},\boldsymbol{b}}(\boldsymbol{x}_i)$ denotes the ith reconstruction of input sample.

The major advantage of AE algorithms lies in the pathway of data dimensionality reduction. They are the efficient data dimensionality reduction methods without supervision. However, the fully connected topology of neural nodes brings obvious limitation to classical AE algorithms. The ignorance of 2-D spatial structure causes it far inferior to CNNs in SAR image classification. Furthermore, the global features obtained by full connection introduce more redundancy in parameter space than those of CNNs. Instead, CNNs models acquire decomposition features and benefit greatly from weights sharing [23], [39]. The 2-D convolutional kernels combined with weights sharing subtly supply local perception fields and focus on spatial locality correlation. And these kernels have already been proved to be more powerful for the representation of multiscale features. In the convolutional operation between two layers, feature maps $\boldsymbol{x}_i^{l-1}$ $(i = 1, 2, \ldots, I)$ of the previous layer are operated by a trainable convolutional kernel. It subsequently generates feature maps of the lth layer. The jth feature map in the next convolutional layer is given by

$$\boldsymbol{x}_j^l = f\left(\sum_{i\in \boldsymbol{M}_j}\boldsymbol{x}_i^{l-1}\cdot\boldsymbol{k}_{ij}^l+\boldsymbol{b}_j^l\right) \tag{3}$$

where $\boldsymbol{x}_j^l$ is the jth feature map of the lth layer, and $f(\cdot)$ denotes an activation function, which greatly enhances models through nonlinear mechanism. $\boldsymbol{M}_j$ stands for the sequence number of associated features. $\boldsymbol{k}_{ij}^l$ denotes a trainable kernel involving the ith input feature map and the jth output feature map. And $\boldsymbol{b}_j^l$ denotes the trainable bias of the jth output feature map.

In general, the implementation of CNN topology requires a considerable amount of supervised input before expressive feedback and classification. On account of this, CAEs are proposed to combine the benefits of both weights sharing and unsupervised learning. They differ from classical AEs as

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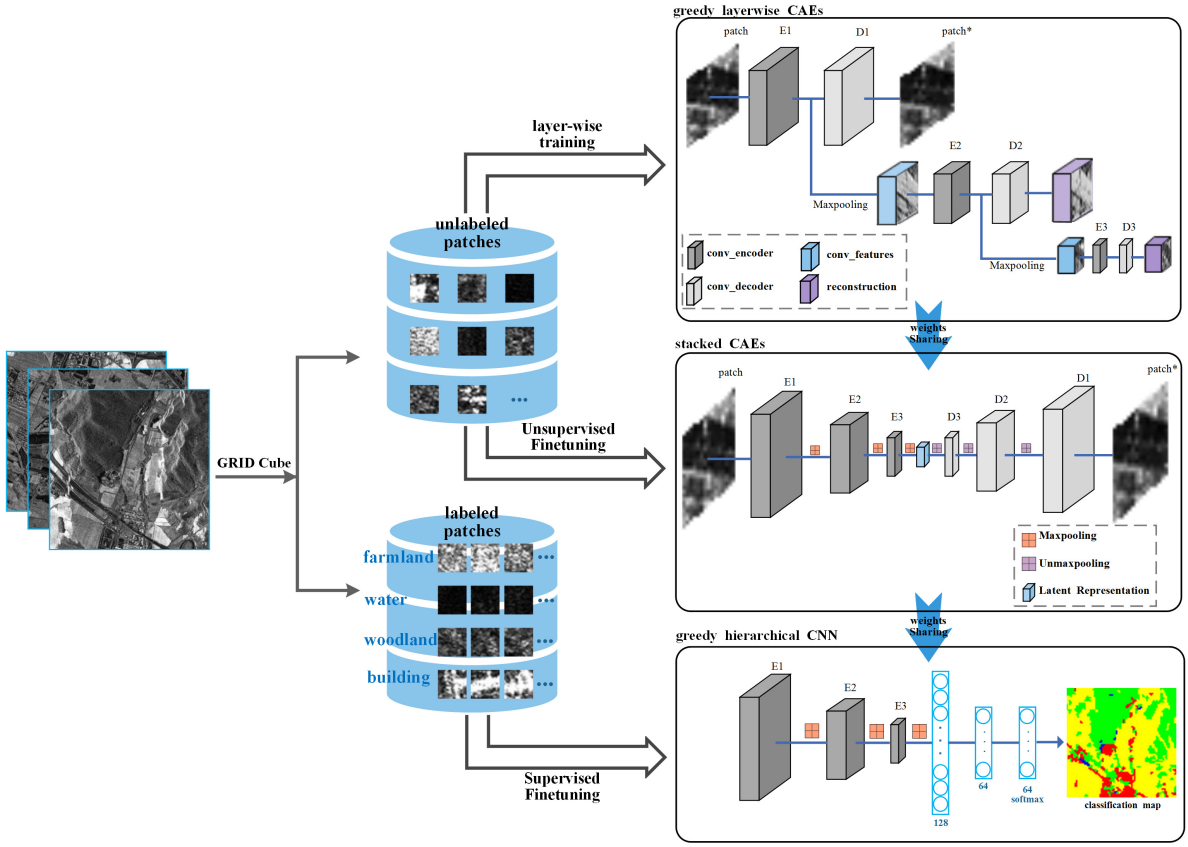


Fig. 1. Overall pipeline for SAR image classification.

their 2-D convolutional topology preserving spatial locality. To some extent, CAE can be regarded as a unit nonlinear transformation of the input signal. For a basic CAE network, including single pair of the convolutional encoder and decoder, the latent representation of the jth feature map h_j is intuitively similar to the form of formula (3). Subsequently, for an input $\boldsymbol{x}$, the reconstruction output $\hat{\boldsymbol{x}}$ derived from the latent feature h is defined as

$$h_j = f\left(\boldsymbol{x} \cdot \boldsymbol{k}_j + \boldsymbol{b}_j\right) \tag{4}$$

$$\hat{\boldsymbol{x}} = f\left(\sum_{j \in \boldsymbol{M}} \boldsymbol{h}_j \cdot \tilde{\boldsymbol{k}}_j + \boldsymbol{c}\right) \tag{5}$$

where $\tilde{\boldsymbol{k}}_j$ denotes antioperation as to the trainable encoding kernel. Formulas (4) and (5) are able to be parameterized by $\boldsymbol{\varpi} = \{\boldsymbol{k}, \boldsymbol{b}\}$ and $\tilde{\boldsymbol{\varpi}} = \{\tilde{\boldsymbol{k}}, \boldsymbol{c}\}$, respectively. The overall mean squared error (MSE) based on formula (2) is the cost function, which aims to achieve an optimal minimization

$$\begin{aligned} \boldsymbol{\varpi}^*, \tilde{\boldsymbol{\varpi}}^* &= \arg\min_{\boldsymbol{\varpi}, \tilde{\boldsymbol{\varpi}}} J(\boldsymbol{x}, \hat{\boldsymbol{x}}) \\ &= \arg\min_{\varpi, \tilde{\varpi}} \frac{1}{n} \sum_{i=1}^{n} \left(\frac{1}{2} \|\hat{\boldsymbol{x}}_i - \boldsymbol{x}_i\|^2\right). \end{aligned} \tag{6}$$

Parameters are then dynamically updated by layer in the process of backpropagation. During this time, these parameters keep chasing after optimization using stochastic gradient descent (SGD).

C. SCAEs With Unsupervised Finetuning

Deeper architecture has great potential to provide more abstract feature synthesis. It has been proved to gain accuracy in many applications [40]. Several CAEs can be coupled together to form a deeper hierarchical structure in a stacked fashion (see Fig. 1). In the stacked cascade, convolutional encoders and decoders are divided to constitute a symmetric topology, which is bounded by the intermediate latent representation. The latent data flow between encoding or decoding layers follows formula (3). Each layer supplies feature abstraction and synthesis as the input of the next layer. Furthermore, the convolutional operation is followed by the maxpooling operation during the encoding process. On the contrary, in the decoding period, the unmaxpooling operation [41] is performed before the convolutional layer. Unmaxpooling is regarded as the inverse operation of maxpooling, which provides a high-efficient approach to map dense representation to a higher spatial dimension.

As depicted in Fig. 2, unmaxpooling with a stride $\boldsymbol{s} = 2$ is an upsampling method, which maps a 2×2 dense feature to 4×4 representation. This operation will double the size of input features. Generally, there are three pathways to perform an unmaxpooling operation. The first one is the nearest neighbor method, which replicates four identical elements to quadruple size. The second approach restores each element value to its initial location by means of relative position records. Replacing each element by an $\boldsymbol{s} \times \boldsymbol{s}$ block with the

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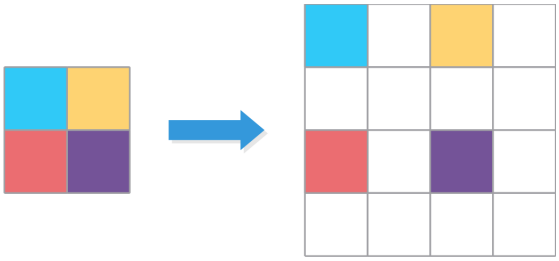


Fig. 2. Diagram of one 2×2 unmaxpooling operation. Unmaxpooling operation is regarded as the inverse of maxpooling by filling kernel fields of unpooled regions with either zeros or copies of maximum element.

element located in the top left corner and zeros elsewhere is another efficient technical measure. This method ensures the accuracy within an acceptable range and considerably reduces the time complexity at the same time. The maxpooling and unmaxpooling can be represented, respectively, as

$$p^{i+1}(\boldsymbol{x}, \boldsymbol{y}) = \max_{(\Delta \boldsymbol{x}, \Delta \boldsymbol{y}) \in \boldsymbol{G}} p^{i}(\boldsymbol{x} \cdot \boldsymbol{s} + \Delta \boldsymbol{x}, \boldsymbol{y} \cdot \boldsymbol{s} + \Delta \boldsymbol{y}) \quad (7)$$

$$p^{i+1}(\boldsymbol{x} \cdot \boldsymbol{s} + \Delta \boldsymbol{x}, \boldsymbol{y} \cdot \boldsymbol{s} + \Delta \boldsymbol{y}) = \begin{cases} p^{i}(\boldsymbol{x}, \boldsymbol{y}), & (\Delta \boldsymbol{x}, \Delta \boldsymbol{y}) = (0, 0) \\ 0, & (\Delta \boldsymbol{x}, \Delta \boldsymbol{y}) \neq (0, 0) \end{cases} \quad (8)$$

where $\boldsymbol{s}$ denotes the stride, and G is the scale of the pooling or unpooling. It contains a series of offset location $(\Delta x, \Delta y)$. In practical use, we usually adopt the scale 2×2 with a stride $\boldsymbol{s} = 2$.

Using algorithm 1 to learn convolutional layers one at a time in an unsupervised greedy strategy, we transform one highly nonconvex optimization into multiple subspaces. Through weights sharing, the layer-wise optimized parameters will be assigned to a complete stacked architecture, which sustains the identical topology as to the single pair of CAEs. It has already been found that the early pretraining has a positive influence on the final global descent procedure [42]. The greedy layer-wise pretraining supplies an unbiased regularizer and *a priori* distribution derived from large volumes of SAR patches for the stacked hierarchy. Let $E(\boldsymbol{y}|\boldsymbol{x})$ be the encoding distribution, which determines a posterior distribution $Q(\boldsymbol{x})$ on the latent data flows. And $P_d(\boldsymbol{x})$ denotes the data distribution. By estimating the probability distribution of training examples, the encoder is also represented as

$$Q(\boldsymbol{x}) = \int_{\boldsymbol{x}} P_d(\boldsymbol{x}) E(\boldsymbol{y}|\boldsymbol{x}) dx \quad (9)$$

where $E(\boldsymbol{y}|\boldsymbol{x}) = E_1(\boldsymbol{y}_1|\boldsymbol{x}) \otimes E_2(\boldsymbol{y}_2|\boldsymbol{x}) \cdots \otimes E_i(\boldsymbol{y}_i|\boldsymbol{x})$ is regarded as the stacked procedure of encoding. Initialization using the greedy algorithm subtly imposes constraints on the overall encoding and decoding distribution, which dramatically benefits the global optimization of the stacked model. After a global unsupervised finetuning, these encoding parameters with comprehensive perception on heterogeneous SAR patches can be used to set an initial stage for a convolutional neural network with identical topology.

D. GHCNN With Supervised Finetuning

Convolutional neural networks transform high-dimensional patterns into many feature maps by the interleaving between multiple connections of convolutional layers and downsampling operations [39]. Subsequently, the dense layers with fully connected topology are involved in a joint classifier. Compared with classical NNs, CNNs are superior to them in three advanced ideas: local receptive fields, shared weights, and subsampling. These innovations, in some ways, enhance the descriptive capability of the model and ensure shift, scale, and distortion invariance. This hierarchical architecture is consistent with the cognitive process, which follows progressive enhancement from the whole to the parts, and from the concrete to the abstract. Furthermore, the self-learning from big data sets and excellent generalization capacity also make convolutional neural networks competitive candidates for SAR image classification.

In general, the architecture of a convolutional neural network consists of four different types of layers: convolution, maxpooling, dense, and softmax. In each convolutional layer, a series of fixed-size kernels is set to carry out convolution operation on SAR images and realize feature extraction. In other words, as for the identical kernel, the signal response of texture structure with random or statistical sense of self-similar yields to an approximation. The data flows of feature maps between two convolutional layers follow

Algorithm 1 Greedy Layerwise CAEs

Input: Training data set $x = (x_1, x_2, \ldots, x_m)$, the number of encoder-decoder pairs L, learning rate ε, training epoch N. Initialize $h^0(x) = x$.

1: **for** $l = 1$ to L **do**
2: Initialize $(\varpi^l, \tilde{\varpi}^l)$, where initializes k^l and $\tilde{k}^l$ by sampling from uniform $(-1, 1)$ randomly, and b^l, c^l are set to 0 temporarily.
3: Define latent features from the lth encoder layer using (4): $h^l(x) = f(k^l \cdot h^{l-1}(x) + b^l)$
4: Define the corresponding decoder layer output using (5): $\hat{x} = f(\sum_{j \in M} k^l_j \cdot h^l_j(x) + c^l)$
5: **while** training epoch $\leq$ N **do**
6: Compute $h^l(x)$ from $h^{l-1}(x)$.
7: Compute reconstruction $\hat{x}$ from $h^l(x)$.
8: Compute cost function $J(h^{l-1}, \hat{x})$.
9: Back-propagation using gradient descent:
compute $\frac{\partial J^l}{\partial \varpi^l}$, where $\varpi^l = (k^l, b^l, c^l)$:

$$\frac{\partial}{\partial k^l} J^l = \frac{1}{m} \sum_{i=1}^{m} \frac{\partial}{\partial k^l_i} J(k, b, c; h^{l-1}_i, \hat{x}_i)$$

$$\frac{\partial}{\partial b^l} J^l = \frac{1}{m} \sum_{i=1}^{m} \frac{\partial}{\partial b^l_i} J(k, b, c; h^{l-1}_i, \hat{x}_i)$$

$$\frac{\partial}{\partial c^l} J^l = \frac{1}{m} \sum_{i=1}^{m} \frac{\partial}{\partial c^l_i} J(k, b, c; h^{l-1}_i, \hat{x}_i)$$

10: Update layer parameters $\varpi := \varpi - \varepsilon \frac{\partial J}{\partial \varpi}$.
11: **end while**
12: **end for**

Output: $\varpi = \{(\varpi^1, \tilde{\varpi}^1), (\varpi^2, \tilde{\varpi}^2), \ldots, (\varpi^L, \tilde{\varpi}^L)\}$.

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Algorithm 2 SCAEs With Unsupervised Finetuning

Input: Training data set $D = (x_1, x_2, \ldots, x_m)$, the number of layers N (N = 2L), learning rate ε, training epoch M, batch number B, which divides the data set into B equal parts $\{d_1, d_2, \ldots, d_B\}$.

Define the function $p(\cdot)$ denotes the maxpooling operation, while the unmaxpooling operation is given as $\tilde{p}(\cdot)$.

Initialize parameter sets of weights and biases $\mu = \varpi$ according to the corresponding relations in algorithm 1, which is parameterized by

$$\{(\mu^1, \mu^{2L}), (\mu^2, \mu^{2L-1}), \ldots, (\mu^L, \mu^{L+1})\} = \{(\varpi^1, \tilde{\varpi}^1), (\varpi^2, \tilde{\varpi}^2), \ldots, (\varpi^L, \tilde{\varpi}^L)\}.$$

while training epoch ≤ M **do**
  for b = 1 to B **do**
    Initialize $x_b^0 = d_b$.
    for i = 1 to L **do**
      Compute $x_b^i = f(x_b^{i-1}; \mu^i)$.
      Compute $O_b^i = p(x_b^i)$.
    end for
    Define $x_b^L = O_b^L$.
    for j = L + 1 to N **do**
      Compute $\tilde{O}_b^j = \tilde{p}(x_b^{j-1})$.
      Compute $x_b^j = f(\tilde{O}_b^j; \mu^j)$.
    end for
    Compute cost function $J_b(x_b^0, x_b^N)$.
    Stacked CAEs parameters update using backpropagation:

$$\frac{\partial J_b}{\partial \mu} := \mu - \varepsilon \frac{\partial J_b}{\partial \mu}$$

  end for
end while

Output: The convergent model of stacked CAEs.

Algorithm 3 GHCNN With Supervised Finetuning

Input: Training data set $D = \{(x_1, y_1), (x_2, y_2), \ldots, (x_m, y_m)\}$, where y denotes the label, the number of convolutional layers L, the number of dense layers F, learning rate ε, training epoch M, batch number B, which divides the data set into B equal parts $\{(X_1, Y_1), (X_2, Y_2), \ldots, (X_B, Y_B)\}$.

Initialize the convolutional parameter space using $\{\mu^1, \mu^2, \ldots, \mu^L\}$ derived from the encoder layers of the SCAEs.
while training epoch ≤ M **do**
  for b = 1 to B **do**
    Initialize $x_b^0 = X_b$.
    for i = 1 to L **do**
      Compute $x_b^i = f(x_b^{i-1}; \mu^i)$ using formula (3).
      Compute $O_b^i = p(x_b^i)$ using formula (7).
    end for
    Define $D_b^0 = O_b^L$.
    for j = 1 to F **do**
      Compute $D_b^j = dense(D_b^{j-1})$.
    end for
    Compute softmax function $\hat{y}_b = softmax(D_b^F)$.
    Compute cost function $C_b(\hat{y}_b, Y_b)$.
    Back- propagation based on gradient $\frac{\partial C_b}{\partial a}$.
  end for
end while

Output: The convergent GHCNN model for SAR image classification.

formula (3). The strategy of local receptive fields restricts the neurons only related to the ones in the fields of the previous layer, which decreases the volume of parameters efficiently. In addition, the maxpooling layer is coupled between two convolutional layers as a joint downsampling operation. As shown in formula (7), this operation is used to achieve dimensionality reduction and boost some degree of distortion invariance. For instance, a 2×2 maxpooling layer with the stride 2 outputs the maximum on the group of neighbor units located within a local 2×2 patch. Based on this, similarities are simplified, and differences are highlighted instead. Moreover, the dense layer is also called fully connected layer. It takes advantage of full-connection topology to map 2-D feature maps to 1-D vectors. Feature transformation on dimensional level introduces the integrated effects on synthesized representation. Then, a softmax layer using these feature vectors realizes nonlinear classification and regression. Softmax function is such a good fit for solving the multiclass pattern recognition of neural networks. The formula is as follows:

$$\hat{\boldsymbol{y}}_i^l = \frac{e^{\boldsymbol{a}_i^l}}{\sum_{j=1}^{\boldsymbol{M}} e^{\boldsymbol{a}_j^l}} \tag{10}$$

where $\boldsymbol{a}^l$ is defined as $\boldsymbol{a}^l = \boldsymbol{k}^T \boldsymbol{x}^l + \boldsymbol{b}$, $\hat{\boldsymbol{y}}_i^l$ denotes the ith output of the lth layer, $\boldsymbol{x}_i^l$ is the ith input of the lth layer, and $\sum e^{\boldsymbol{a}_j^l}$ stands for the weighted sum of all neurons derived from the lth layer. $\sum e^{\boldsymbol{a}_j^l}$ can be regarded as a normalized term. In general, cross entropy is introduced as the cost function to evaluate the performance of the classification model. The cross entropy function is defined as

$$C(\boldsymbol{y}, \hat{\boldsymbol{y}}) = -\sum_{j=1}^{\boldsymbol{M}} \boldsymbol{y}_j \ln \hat{\boldsymbol{y}}_j \tag{11}$$

where $\boldsymbol{y}_j$ denotes the label value of the jth neuron, which has been declared in the training data set. Similar to AEs, the gradient of the cost function with respect to the parameter $\boldsymbol{a}_i$ can also be computed based on the backpropagation mechanism [43]. The gradient of the cost function is defined as

$$\frac{\partial C}{\partial \boldsymbol{a}_i} = \frac{\partial C}{\partial \hat{\boldsymbol{y}}_i} \frac{\partial \hat{\boldsymbol{y}}_i}{\partial \boldsymbol{a}_i} + \sum_{j \neq i}^{\boldsymbol{M}} \frac{\partial C}{\partial \hat{\boldsymbol{y}}_j} \frac{\partial \hat{\boldsymbol{y}}_j}{\partial \boldsymbol{a}_i} \tag{12}$$

where the derivative of the cost function is divided into two cases depending on i and j. When $i = j$, the equation can be extracted as

$$\frac{\partial C}{\partial \hat{\boldsymbol{y}}_i} \frac{\partial \hat{\boldsymbol{y}}_i}{\partial \boldsymbol{a}_i} = -\frac{\boldsymbol{y}_i}{\hat{\boldsymbol{y}}_i} \left[\hat{\boldsymbol{y}}_i (1 - \hat{\boldsymbol{y}}_i)\right]. \tag{13}$$

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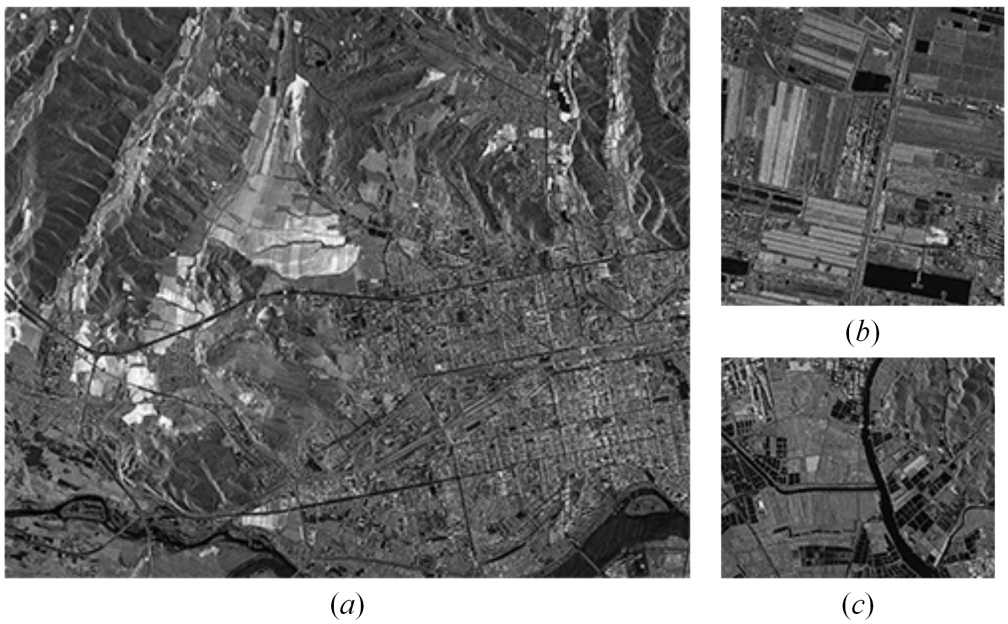


Fig. 3. GF-3 SAR data sets for three different experimental regions. (a) Region A: Mudanjiang, Northeast China. (b) Region B: Dongying, East China. (c) Region C: Zhongshan, South China.

Otherwise, the partial differential equation is

$$\sum_{j\neq i}^{M}\frac{\partial C}{\partial \hat{\boldsymbol{y}}_j}\frac{\partial \hat{\boldsymbol{y}}_j}{\partial \boldsymbol{a}_i}=\sum_{j\neq i}^{M}\left[-\frac{\boldsymbol{y}_j}{\hat{\boldsymbol{y}}_j}\left(-\hat{\boldsymbol{y}}_i\hat{\boldsymbol{y}}_j\right)\right]. \tag{14}$$

Based on the SGD algorithm, the weights of the CNN will be updated iteratively in a supervised manner.

The training trajectory of NN models attaches much importance on the initialization of parameter space and the selection of optimization algorithm. In the benchmark process of training CNNs, the widespread adoption of random initialization on parameter space tends to cause poor generalization, or even make it fall into many issues such as local minima, overfitting, gradient diffusion [44], and so on. To tackle with these issues, we can take measures to enhance the robustness of cost function in terms of imposing regularization throughout training [45], or deploy a strategy called "dropout" to control the activated state of neurons [46]. Meanwhile, the greedy hierarchical learning is originally applied to the identical convolutional topology in a separated semisupervised pipeline. It significantly speeds up the training process and achieves superior convergence with regard to the nonconvex constrained optimization. In the patch-based classification, the greedy hierarchical model is proposed to inherit the parameter space derived from pretrained encoders of the SCAEs. Afterward, several dense layers and a softmax layer are integrated into the GHCNN model toward a global supervised finetuning.

III. Experimental Results

This section deeply analyzes and demonstrates the proposed method and its technical maturity based on a wide diversity of regional instances and parameter setup. In Section III-A, the data sets of three different experimental regions are presented in detail. Section III-B discusses the selection process of optimal parameters. Visual efficiency analysis for the proposed greedy hierarchical algorithm is introduced in Section III-C. Section III-D evaluates the accuracy of the proposed and several contrast methods.

A. Data Sets

The data sets used in terrain classification experiments are from GF-3 satellite, which carries a C-band SAR with multiple imaging modes. The SAR data sets of three different experimental regions are shown in Fig. 3(a)–(c). All of the selected images are from spotlight mode with an identical ground sampling distance (GSD) 1 m $\times$ 1 m after a series of standardized correction, calibration, geocoding, and resampling process. In Fig. 3, the region A data set with a size of 11 520 $\times$ 9600 pixels is located in Mudanjiang, Northeast China. This region is on behalf of typical northern urban areas in China. The scene of this SAR image not only covers large areas of building, farmland, water, and woodland, but also the messy mixing of the four abovementioned land terrain types. This kind of large inhomogeneous area requires a higher algorithm performance for SAR image classification. Region B is situated in Dongying, East China. And the region C comes from Zhongshan, South China. The coverage areas are 4968 $\times$ 5232 and 4800 $\times$ 3864 pixels for SAR, respectively. Both of them contain four types of land cover: building, farmland, water, and woodland. These two areas are on behalf of the typical east plain and south hilly area in China. Other key parameters of the satellite platform and radar sensor are listed in Table I, including polarization, angle, orientation, and so on.

B. Parameter Setup for Proposed Approach

The selection process of hyperparameters has a far-reaching influence on image classification using NNs [47]. To a large extent, it determines the performance of models in the aspect

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TABLE I
PARAMETERS OF SATELLITE PLATFORM AND SAR DATA

Region	Size	Polarization	Orbit Direction	Antenna Direction	Beamforming Angle	Receipt Time
A	11520×9600	HH	ASC	R	26.31°	20170514UTC
B	4968×5232	VV	ASC	R	34.70°	20170416UTC
C	4800×3864	HH	ASC	R	26.31°	20161201UTC

of representation, velocity and veracity. In our experiments, multiple hyperparameters are evaluated on multiscale trajectories for the given data sets, including the size of SAR patches, the depth of greedy pretrained layers, the number of features, and the receptive field size of convolution kernels [48]. The CAEs are trained on the data sets without labeled identification in a greedy hierarchical approach. By this way, the original distribution of SAR images is sensed and set for the disordered convolutional parameter space with an unsupervised strategy. We then use these pretrained layers to finetune an SAEs model based on several weights sharing operators to acquire a global homeostasis model efficiently. Afterward, a convolutional neural network with the same topology as encoding layers is constructed for SAR image classification. It adopts weights sharing operators and fully connected layers to implement a supervised finetuning on the identical labeled data sets. Our experiments are conducted by the scientific python development environment Spyder[1] on the platform of Windows with Intel Core i7-4790K CPU, NVIDIA Tesla K20c GPU, and 32-GB memory. In addition, all algorithms use an end-to-end open-source machine learning framework Tensorflow[2] and a parallel computing library CUDA.[3]

In this section, all experiments and analyses for the hyperparameters setup implement on the three given SAR data sets. These data sets contain large coverage and messy mixing areas with typical representativeness, which put forward higher requirements on the algorithm performance. The optimal parameters on these three data sets are subsequently presented in the follow-on analyses, respectively. We apply overall accuracy and kappa coefficient as the global evaluation criteria for SAR image classification by fivefold cross-validation method.

As for the patch size setup, we set a series of scales ranging from 16 to 32 with fixed offsets (see Fig. 4). It is intuitively illustrated in Fig. 4 that varying scales of SAR patches cause a slight fluctuation of classification accuracy within a certain range. As the scale increasing, the overall accuracy (OA) results of different data sets all take on the single peak distributions. The optimal patch scale is located at 24, where all of the three regions achieve the best performance. The appropriate scale reaches a balance between the degree of discrimination and confusion for SAR image classification. If it is below the optimal scale section, the patch is too small to catch enough features. The oversize scale introduces lots of confusion instead. At the same time, it is presented that the region size and mixing of various land types have significant impacts on accuracy. As for region C, it obtains the best

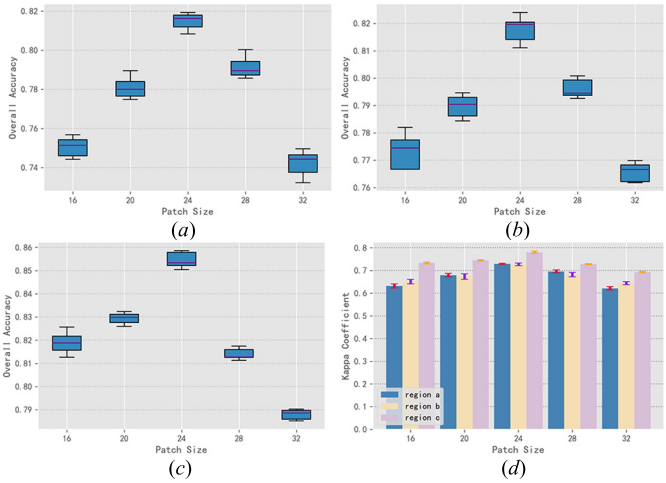


Fig. 4. Statistical results of OA and kappa coefficient varying with a series of given patch sizes for the three data sets. (a)–(c) OA of the Mudanjiang, Dongying and Zhongshan data sets given multiscale patch sizes. (d) Kappa coefficient with error bars of the three data sets given different patch sizes.

classification accuracy with a high degree of consistency [see Fig. 4(d)]. This region focuses on the minimal coverage with large homogeneous area in the three data sets.

In the second setup, we discuss the influence of variant GH layer settings by training networks with different weights sharing operators. The depth of greedy pretrained layers is set ranging from 0 to 3 (see Fig. 5), which determines the number of layers with unsupervised greedy hierarchical pretraining. Fig. 5(a) depicts a decline regularity of classification error on three given data sets as the number of GH layers increasing. This indicates that the greedy hierarchical learning contributes to the complex nonconvex optimization for SAR image classification task. For instance, the sample curves of training loss on the region A data set show that the network with fully hierarchical pretraining is better at generalization within normal epochs, compared with no hierarchical pretraining or top-layer pretraining [see Fig. 5(b)]. The volumes of training samples of three data sets are shown in Table II. At the same time, it is noted that the classification error is sensitive to the volume of data sets. The variable range of accuracy exceeds 10% for the first region with the depth of greedy pretrained layers increasing. Instead, it is kept within 5% for region C. We can draw the conclusion that our method achieves even better efficiency on the large volume of SAR data sets.

After exploring the effect of depth, in the third setup, we carry out a series of experiments to probe the width of neural networks with a number of incremental kernels in three convolutional layers. Convolutional kernels represent the number of features in each layer, which determine diversified

[1]https://pypi.org/project/spyder/
[2]https://tensorflow.google.cn/
[3]https://developer.nvidia.com/cuda-toolkit/

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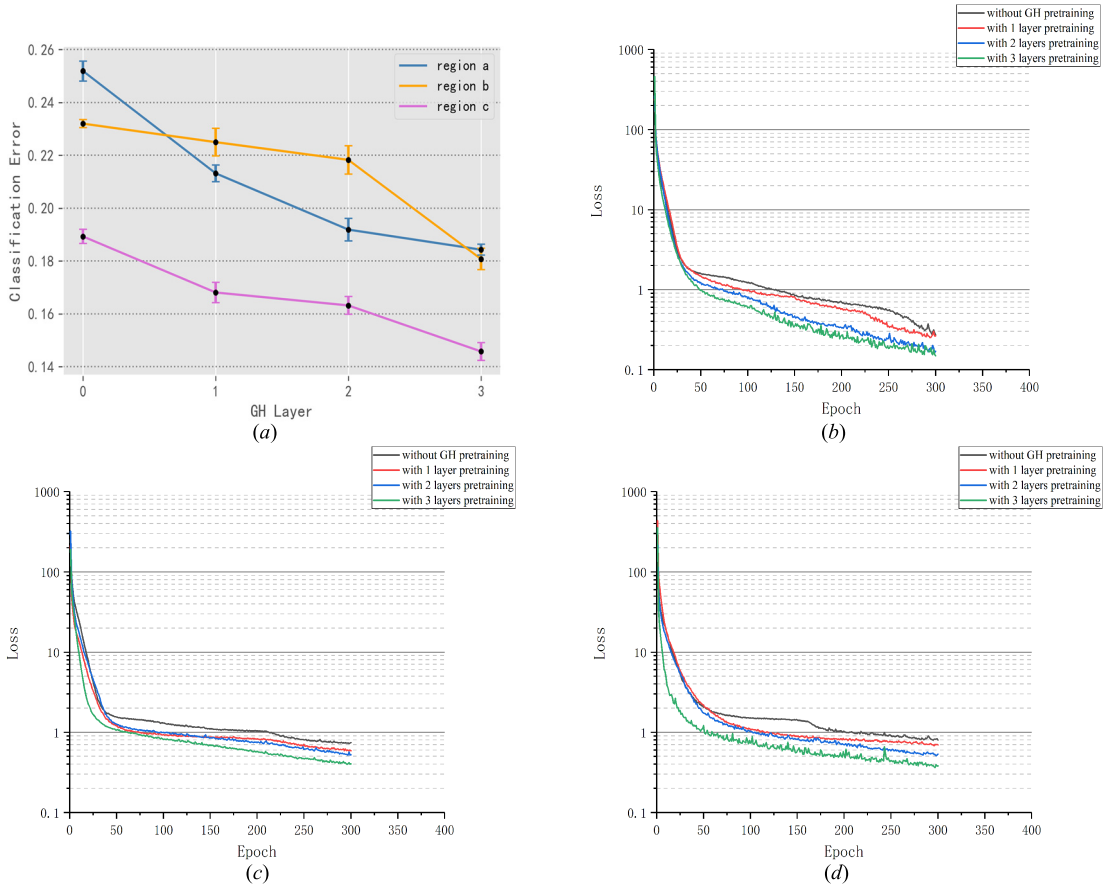


Fig. 5. Comparison among varying GH-layer pretraining for three data sets. (a) Classification errors on three data sets given different GH layers. (b)–(d) Sample curves of training loss on region A–C data sets, respectively.

TABLE II
TRAINING SAMPLES OF THREE DATA SETS

Region	Supervision				Identical no-supervision
	building	farmland	water	woodland	
A	6000	6000	6000	6000	24000
B	4000	4000	4000	4000	16000
C	2000	2000	2000	2000	8000

TABLE III
OA GIVEN DIFFERENT KERNEL SETTINGS

Dataset	16-8-4	64-32-16	256-128-64
A	60.67±0.56	80.53±0.50	**81.57±0.20**
B	67.34±0.35	**81.91±0.40**	78.02±0.67
C	84.63±1.03	**85.41±0.33**	80.70±0.26

hierarchy representation. In this case, we design three architectures using varying kernel settings throughout the whole classification pipeline. The classification results are presented in Table III. All experiments are performed with a fixed epoch setting under the premise of global convergence. Intuitively, it is observed that the different kernel settings are unable to achieve fully consistent performance on three data sets. As for the region A, our classification model gets best performance on the third-level setting 256-128-64, while regions B and C all achieve best performance on the second scale. On the first setting, regions A and B appear worst results on classification accuracy, and region C shows a good result instead. It should be noted that we focus on the single-polarized SAR images, which contain the texture as the main form of discriminative information. As for the complex images with large coverage, a relatively modest increase in convolutional kernels has a good effect on promoting classification accuracy. However, it is proved that the overgrowth of kernels introduces excess redundancy and self-correlation between feature maps in each layer, which causes inefficient training and overfitting.

The next step is to analyze the receptive field size of convolutional kernels. Partial experimental results are displayed in Fig. 6. In this section, we discuss the effect of kernels with different receptive fields on the whole classification pipeline. Traditional 3×3 and 5×5 kernels are tested in our experiments. And at the same time, dilated convolution kernels [49] are also introduced to explore the impact of the novel perceptive mode. As we can see in Fig. 6(a) and

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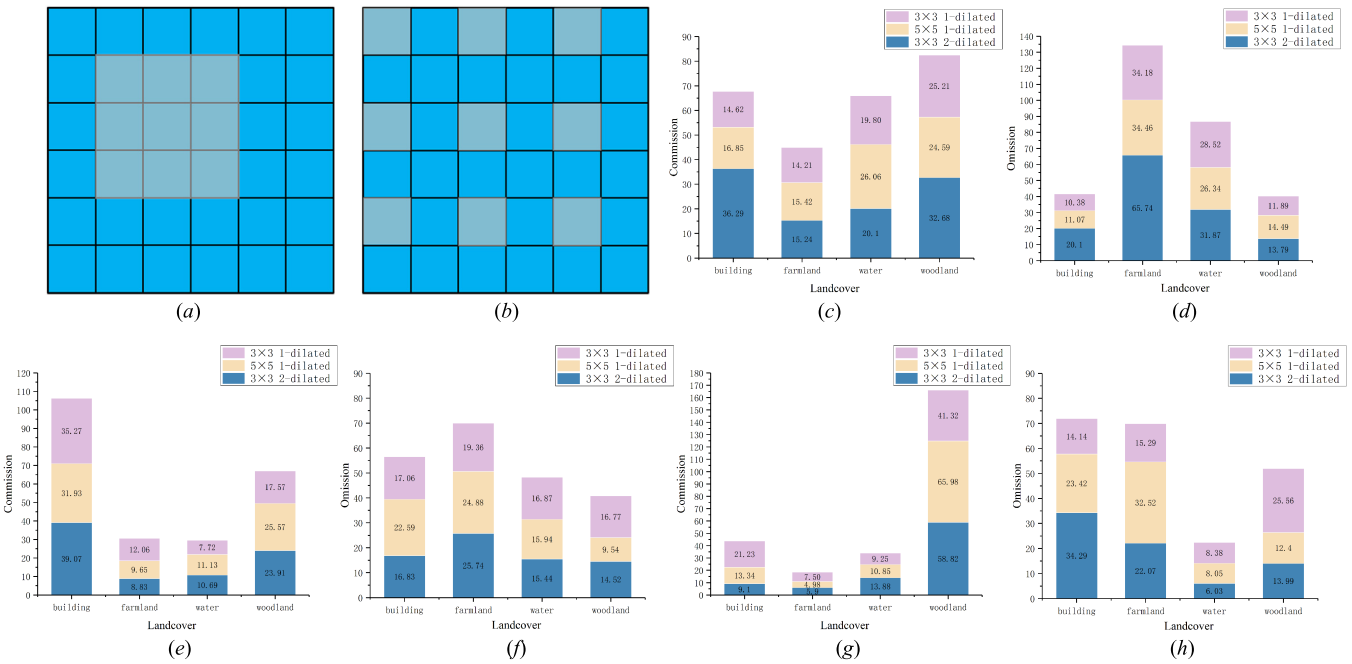


Fig. 6. Classification performance comparison on convolutional kernels with different receptive fields. (a) 3 × 3 kernel with 1-dilated rate. (b) 3 × 3 kernel with 2-dilated rate. (c), (e), and (g) Classification commission displayed as stacked bars on region A–C data sets, respectively. (d), (f), and (h) Classification omission displayed as stacked bars on region A–C data sets, respectively.

TABLE IV
CONVOLUTIONAL NETWORKS CONFIGURATIONS

Greedy layerwise CAEs	Stacked CAEs	GHCNN Configuration
Conv. 64/128@[3,3] Conv. 64/128@[3,3]	Conv. 64/128@[3,3]/CAEs weights sharing Max Pooling [2,2]-s=2 Conv. 32/64@[3,3]/CAEs weights sharing Max Pooling [2,2]-s=2 Conv. 16/32@[3,3]/CAEs weights sharing Max Pooling [2,2]-s=2 Hidden Representation Unmaxpooling Conv. 16/32@[3,3]/CAEs weights sharing Unmaxpooling Conv. 32/64@[3,3]/CAEs weights sharing Unmaxpooling Conv. 64/128@[3,3]/CAEs weights sharing	Conv. 64/128@[3,3]-ReLU/SCAEs weights sharing Max Pooling [2,2]-s=2 Conv. 32/64@[3,3]-ReLU/SCAEs weights sharing Max Pooling [2,2]-s=2 Conv. 16/32@[3,3]-ReLU/SCAEs weights sharing Max Pooling [2,2]-s=2 Dense. 128-ReLU-Dropout Dense. 64-ReLU-Dropout Output. -Softmax
Max Pooling (encoder 1) Conv. 32/64@[3,3] Conv. 32/64@[3,3]		
Max Pooling (encoder 2) Conv. 16/32@[3,3] Conv. 16/32@[3,3]		

(b), the 3 × 3 kernel with a 2-dilated rate injects holes in the compact convolution map, which gains a part of 5 × 5 receptive field. For example, on the region A data set, the classification commission and omission are depicted in Fig. 6(c) and (d) as stacked bars, in which the inconsistent discrimination of different land cover types is revealed with varying receptive fields. It deserves special notice that the 3 × 3 kernels appear to be effective in patch-based SAR image classification. And the dilated convolution kernels perform poorly in most land cover types. However, it can be found that large-scale kernels have advantages in the extraction of building and woodland. That is to say, larger receptive fields achieve a better perception of inhomogeneous texture features.

C. Efficiency Analysis of Greedy Hierarchical Method

In the previous experiments, we figure out a series of optimal parameters for the proposed classification framework by varying multiple hyperparameters. The optimal hyperparameters and topological structure of the proposed model are listed in Table IV. In order to further probe into the superiority and efficiency of the GHCNN model, we firstly design visualization experiments at the convolutional feature level on the basis of above work. And in the second part, stepwise analytic models with different configurations are also taken into consideration. Multiple separate combinations of the greedy hierarchical model are analyzed, respectively, in hundreds of repeated experiments.

In the first part, the early stage and late-stage training process of GHCNN and baseline CNN are under careful surveillance. Intuitively, partial experimental results are illustrated in Fig. 7. As for the given data sets, the loss function curves of two models are depicted in Fig. 7(a), (c), and (e). These results show a high degree of consistency that greedy hierarchical models possess substantial advantages on efficiency and accuracy over baseline models. Moreover,

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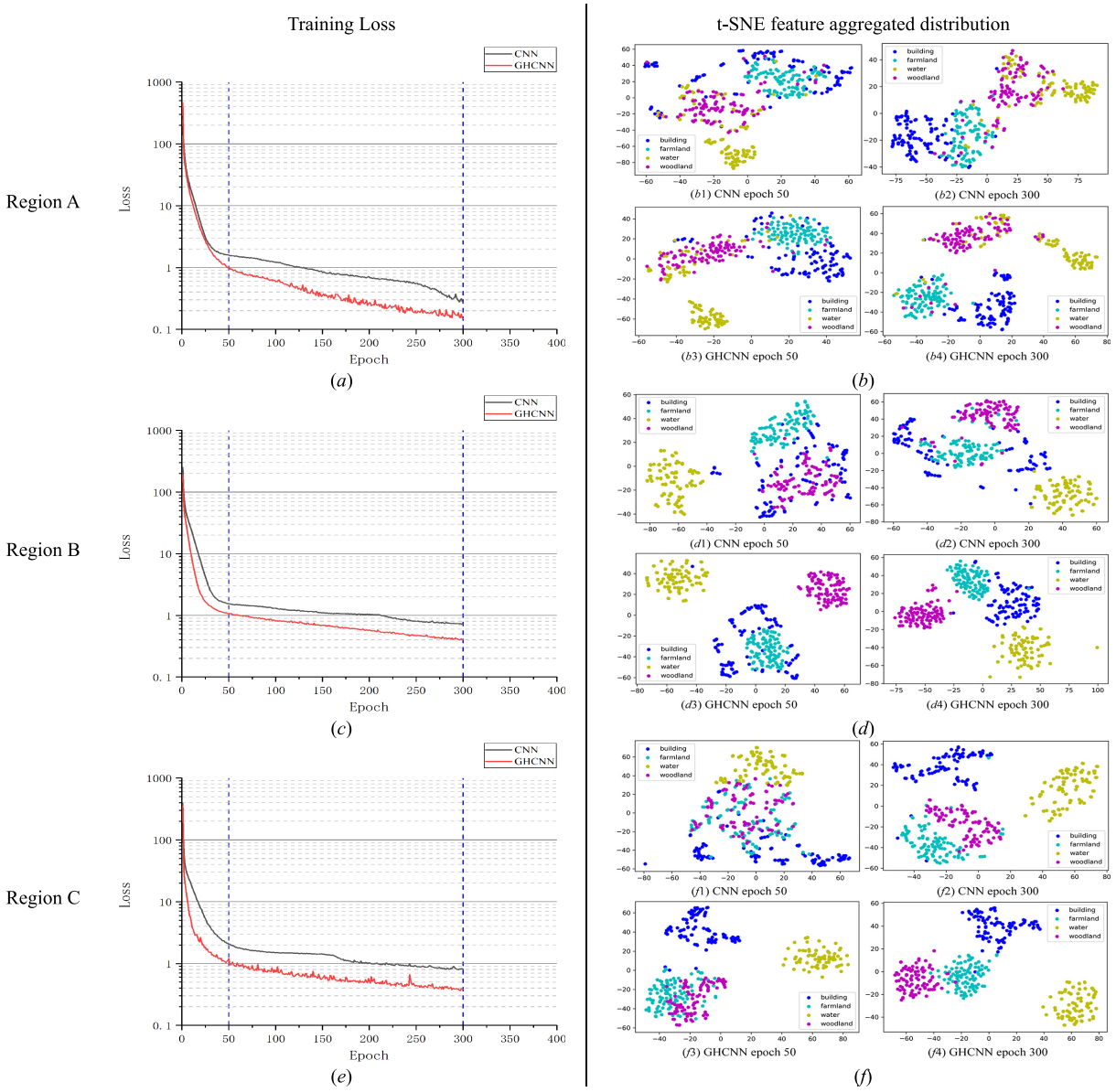


Fig. 7. Comparison of training loss and t-SNE feature aggregated distribution between CNN and proposed GHCNN models on three data sets. (a), (c), and (e) Training loss of CNN and GHCNN models after 300 epochs. (b), (d), and (f) Feature aggregated distribution of four land cover types with 100 random samples for each class using t-SNE algorithm in the 50th and 300th epoch.

a mass of experimental results also demonstrates the theory basis in Section II that the models with greedy hierarchical pretraining tend to accelerate convergence and solve complex nonconvex optimization problems. In general, the GHCNN models are more liable to acquire lower loss function after dozens of epochs, which represents a better parameter space and nonlinear solution.

In the meantime, t-distributed stochastic neighbor embedding (t-SNE) [50] algorithm is applied to evaluate the intermediate feature maps of these black-box models. It has been widely proved that t-SNE is an efficient nonlinear dimensionality reduction algorithm for high-dimensional data mining. Visualization t-SNE algorithm is able to map multidimensional data to 2-D points, which supports for ease of visual analysis. As for convolutional neural networks, feature maps are the core intermedium of data stream between convolution layers. In order to grasp the difference in the efficiency of GHCNN and baseline CNN models, our experiments focus on the algorithm performance of feature discriminability during the training process. According to the optimal topology of the proposed models, we select the bottom convolution layer with advanced abstraction as an analysis example. The tensor structure of multiple feature maps extracted from the third GHCNN layer is a data cube with $6 \times 6 \times 16$ 3-D space. This data cube represents a collection of 16 different feature maps, which retains the essence of a certain land cover type. As is shown in Fig. 7(b), (d) and (f), we randomly select 400 SAR patches with 100 patches per type to analyze the feature aggregated distribution in the 50th and 300th epoch. It is observed that the GHCNN model has the capacity to achieve even superior feature recognition in the early training stage compared with the baseline CNN model. With the loss function decreasing,

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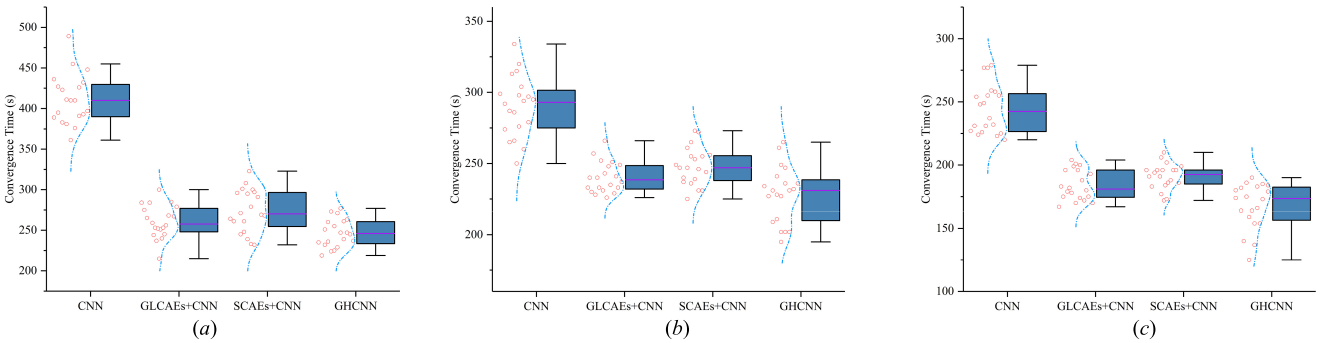


Fig. 8. Stepwise efficiency analysis of the proposed model on three data sets. (a)–(c) Initial convergence efficiency of separate combined models obtained by 20 Monte Carlo runs, respectively, on region A–C data sets.

TABLE V
HYPERPARAMETER SETTINGS OF CONTRAST METHODS

Algorithm	Hyper-parameter Settings
SVM	C-SVC = 0.6, kernel = 'poly', degree = 3, gamma = 'auto', coef0 = 0.0, class_weight = None, Monte Carlo random_state = True.
RF	n_estimators = 500, criterion = 'gini', max_features = 'auto', min_samples_leaf = 1, bootstrap = True, oob_score = True, Monte Carlo random_state = True.
GBDT	base_estimator = DecisionTreeClassifier, n_estimators = 100, algorithm = 'SAMME.R', Monte Carlo random_state = True.
SCAE	lr = 0.01, batch_size = 50, epoch = 100, loss function = 'MSE', optimizer = 'Adam', data shuffle = True.
SDAE	lr = 0.01, batch_size = 50, epoch = 100, activation = 'Sigm', masknoise = 0.1, dropout = 0.1, nonSparsityPenalty = 0.1, loss function = 'Cross Entropy', optimizer = 'SGD', data shuffle = True.
Alexnet	pretrained model = True, patch resize = True, lr = 0.001, batch_size = 256, finetuning epoch = 40, activation = 'Relu', dropout = 0.4, loss function = 'Cross Entropy', optimizer = 'SGD', data shuffle = True.
Caffenet	pretrained model = True, patch resize = True, lr = 0.001, batch_size = 256, finetuning epoch = 40, activation = 'Relu', dropout = 0.4, loss function = 'Cross Entropy', optimizer = 'SGD', data shuffle = True.
VGG-S	pretrained model = True, patch resize = True, lr = 0.001, batch_size = 256, finetuning epoch = 15, activation = 'Relu', dropout = 0.4, loss function = 'Cross Entropy', optimizer = 'SGD', data shuffle = True.
VGG-16	pretrained model = True, patch resize = True, lr = 0.001, batch_size = 64, finetuning epoch = 10, activation = 'Relu', dropout = 0.4, loss function = 'Cross Entropy', optimizer = 'SGD', data shuffle = True.
CNN	lr = 0.001, batch_size = 256, epoch = 300, activation = 'Relu', dropout = 0.4, loss function = 'Cross Entropy', optimizer = 'Adam', data shuffle = True.

the multiple types of SAR patches are gradually identified clearly in the t-SNE distribution maps. In the 300th epoch, more significant difference in the degree of aggregation is revealed obviously. The feature maps extracted from the proposed models can be classified easily using linear or nonlinear classifiers. However, the clutter and error distributions still exist in the baseline CNN distribution maps obviously.

In the second part, a stepwise analysis of the proposed model is carried out for further verification. We design hundreds of repeated experiments to analyze the contribution of each module. Four models are constructed in a standalone approach, including baseline CNN without greedy learning, CNN with greedy layer-wise CAEs pretraining, CNN with SCAEs pretraining, and our proposed model. For each model, we carry out 20 Monte Carlo runs on three given data sets, respectively.

The convergence time of repeated runs is depicted in the statistical graphs. In Fig. 8(a)–(c), the curve fitting and boxplots of the statistical results are also presented, respectively. It can be intuitively observed that the distribution of convergence time is greatly affected by the model structure. Greedy learning algorithm brings about huge improvements to the efficiency of baseline CNN model. The nonconvex optimization of complex SAR pattern recognition benefits a lot from the parameter-oriented guidance.

D. Classification and Comparison

In this section, we design a multiset of contrast experiments based on several widely accepted algorithms using the given SAR data sets. To verify the validity and effectiveness of our proposed algorithm framework, three non-NN machine

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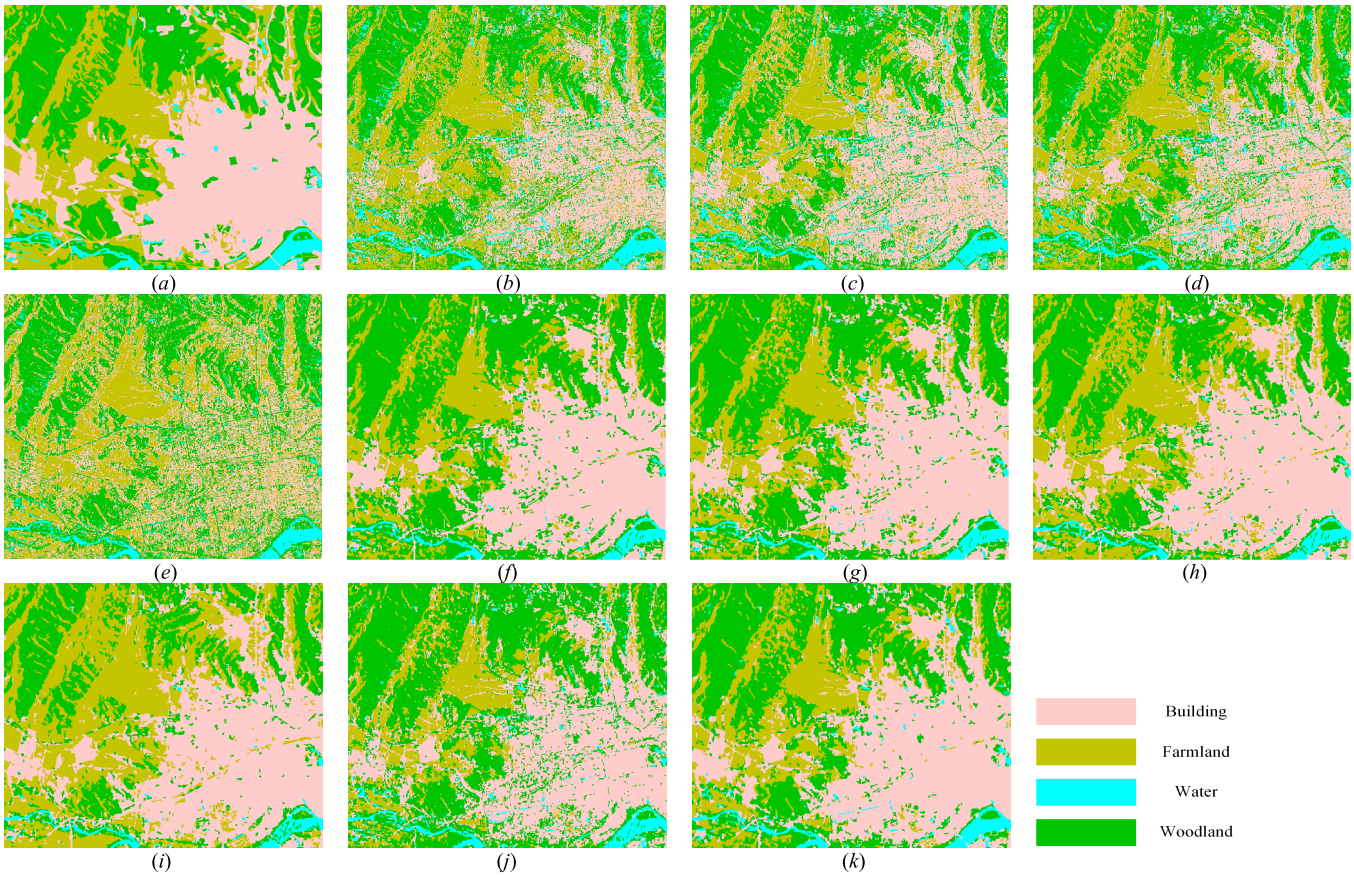


Fig. 9. Classification maps of involved algorithms on the MDJ data set. (a) Ground truth. (b) SVM. (c) RF. (d) GBDT. (e) SDAE. (f) Alexnet. (g) Caffenet. (h) VGG-S. (i) VGG-16. (j) CNN. (k) GHCNN.

learning algorithms and seven NN deep learning algorithms are adopted to conduct controlled experiments, including SVM [51], RF [52], GBDT [53], SCAE [54], SDAE [55], baseline CNN, Alexnet [23], Caffenet [56], VGG-S [57], and VGG-16 [58]. All algorithms in our experiments use the same SAR data sets as inputs. Furthermore, the Monte Carlo random state and data shuffle strategy are introduced to guarantee the independence of multiple cross-validation experiments. The hyperparameter settings of different contrast algorithms are listed in Table V. The reasonable adjusting of parameters is performed to ensure the convergence and efficiency of involved classification models.

As for accuracy evaluation indexes, we take advantage of producer's accuracy (PA), user's accuracy (UA), OA, and kappa coefficient. PA indicates the fraction of correctly classified pixels with regard to all pixels of that ground truth class, while UA represents the fraction of correctly classified pixels with regard to all pixels classified as this class in the classified image. OA is calculated as the total number of correctly classified pixels (diagonal elements in confusion matrix) divided by the total number of test pixels. The kappa coefficient measures the agreement between the classification map and ground truth value, which ranges from 0 to 1. A kappa value of 1 represents a perfect agreement, while a value of 0 shows no agreement. Moreover, the total elapsed time of different algorithms is also used as an important computational efficiency evaluation.

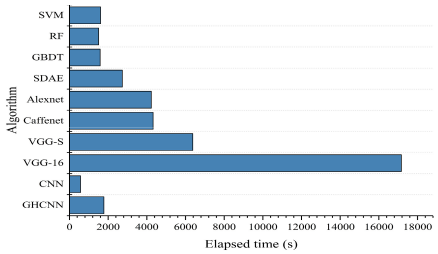


Fig. 10. Total elapsed time of entire classification procedure using different algorithms on region A data set.

1) Classification Experiments on the MDJ Data Set: It can be observed in Fig. 3 and Table I that the MDJ data set covers the largest area with high complexity in three given data sets. A set of optimal parameters for the proposed and contrast algorithms is deployed completely according to Section III-B and Table V. The training strategies of 5 Monte Carlo random state and data shuffle contribute to duplicating the experiment multiple times and figuring out statistically significant results. The comparison of classification maps, elapsed time, and accuracy results is displayed in Figs. 9 and 10 and Table VI.

According to Table VI, what is noticeable is that deep models achieve a better performance than conventional non-NN models in terms of most land cover types. Especially for building and woodland, deep models can increase

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TABLE VI
ACCURACY COMPARISON OF CLASSIFICATION ON MDJ DATA SET

Algorithm	PA				UA				OA	Kappa
	Building	Farmland	Water	Woodland	Building	Farmland	Water	Woodland		
SVM	62.84±0.19	63.72±0.16	76.57±0.19	74.94±0.10	75.64±0.05	68.70±0.18	46.61±0.11	62.02±0.12	67.19±0.09	0.5276±0.0012
RF	71.58±0.11	56.83±0.19	**78.00±0.13**	80.07±0.05	72.93±0.10	78.49±0.11	47.12±0.07	65.76±0.15	70.02±0.08	0.5655±0.0012
GBDT	66.72±0.21	62.83±0.31	77.39±0.10	82.67±0.12	80.70±0.17	74.78±0.13	46.82±0.10	63.71±0.04	70.71±0.06	0.5784±0.0009
SDAE	46.19±0.02	58.62±0.01	66.99±0.04	72.04±0.32	59.34±0.18	58.49±0.28	57.05±0.41	57.42±0.12	58.27±0.10	0.3969±0.0014
Alexnet	85.68±0.10	61.33±0.26	57.59±0.08	**91.82±0.07**	**87.29±0.09**	86.66±0.12	91.45±0.11	67.95±0.13	79.43±0.05	0.6977±0.0007
Caffenet	88.14±0.36	62.06±0.13	60.15±0.25	89.77±0.44	84.29±0.10	87.04±0.21	89.96±0.19	69.33±0.22	79.44±0.26	0.6993±0.0028
VGG-S	88.80±0.04	74.33±0.09	59.53±0.05	80.07±0.05	84.84±0.03	78.03±0.08	89.25±0.01	78.03±0.03	80.95±0.01	0.7196±0.0001
VGG-16	87.37±0.12	**82.56±0.12**	52.64±0.08	65.53±0.18	85.84±0.13	67.39±0.12	**92.16±0.04**	**81.21±0.12**	78.23±0.04	0.6798±0.0006
CNN	80.35±0.71	51.36±0.17	71.83±0.18	91.15±0.55	81.91±0.54	**89.34±0.63**	72.59±1.64	63.27±0.12	74.80±0.37	0.6315±0.0053
GHCNN	**89.62±0.85**	65.81±0.24	71.47±0.91	88.10±0.11	85.38±0.38	85.79±0.80	80.19±0.52	74.79±0.37	**81.57±0.20**	**0.7295±0.0028**

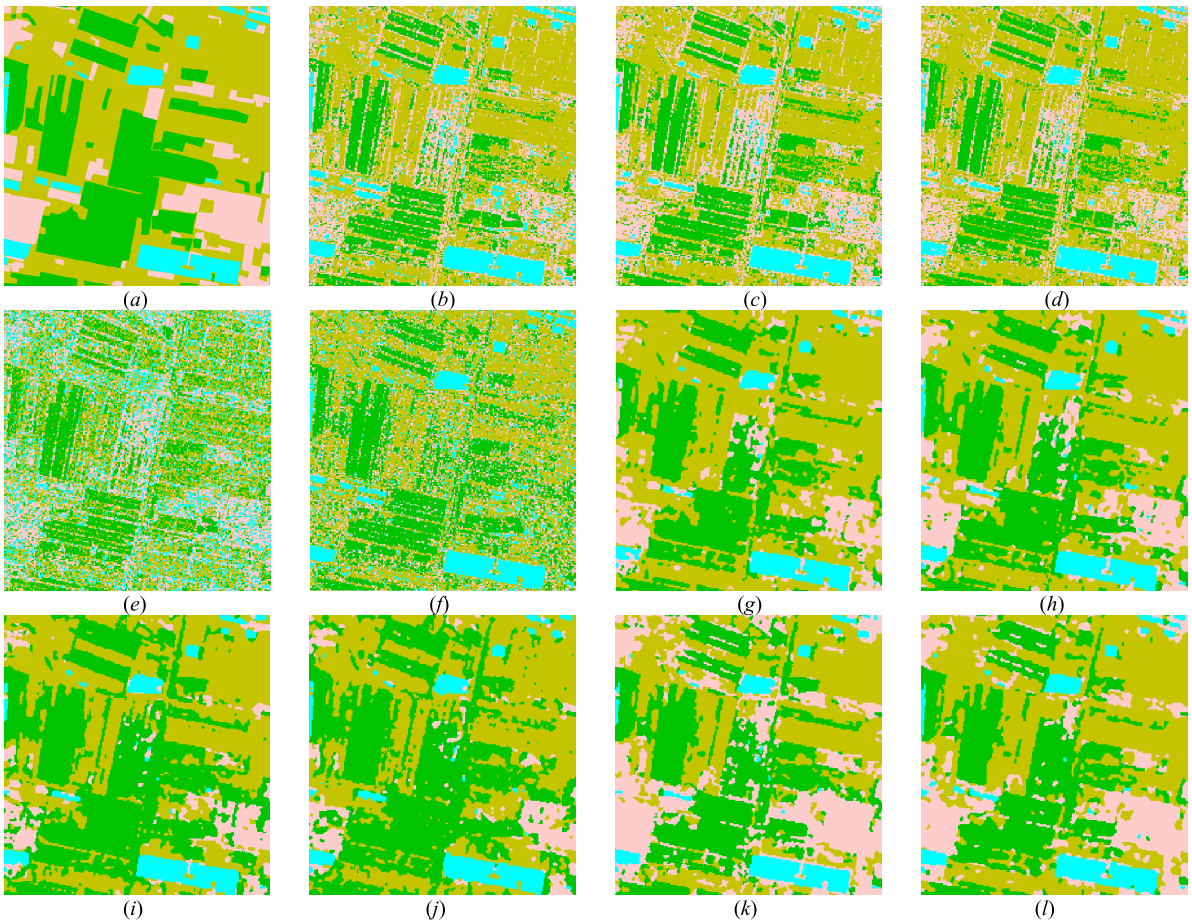


Fig. 11. Classification maps of involved algorithms on the DY data set. (a) Ground truth. (b) SVM. (c) RF. (d) GBDT. (e) SCAE. (f) SDAE. (g) Alexnet. (h) Caffenet. (i) VGG-S. (j) VGG-16. (k) CNN. (l) GHCNN.

classification accuracy by roughly 10%. The OA and kappa coefficient of convolutional models are also accounted for the absolute superiority. However, it is worth noting that the non-NN models have an advantage in water extraction over NN models in terms of the PA. We can infer that multilevel convolution features have low discrimination sensitivity to water, which contains the simple texture with homogeneous low intensity. As for the proposed GHCNN model, it achieves the overall superiority to the baseline CNN model. This demonstrates that the greedy hierarchical pretraining brings significant benefits in model optimizing during the same finite epochs. Compared with other deep models, the proposed model achieves a better accuracy in building and water extraction. The VGG-S model also shows a pretty good result. The total elapsed time of different algorithms is shown in Fig. 10. It can be observed that the deeper models cost a huge amount of computational resources to complete the whole classification pipeline. In the

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TABLE VII
ACCURACY COMPARISON OF CLASSIFICATION ON DY DATA SET

Algorithm	PA				UA				OA	Kappa
	Building	Farmland	Water	Woodland	Building	Farmland	Water	Woodland		
SVM	52.31±0.70	73.60±0.13	**87.49±0.03**	56.66±0.62	32.61±0.24	78.33±0.13	64.47±0.22	80.42±0.47	66.16±0.13	0.4974±0.0013
RF	64.39±0.29	72.67±0.17	83.42±0.07	62.47±0.21	38.22±0.27	80.82±0.12	80.60±0.19	78.29±0.24	69.05±0.10	0.5408±0.0010
GBDT	58.09±1.03	78.83±0.26	76.63±0.39	59.87±0.09	40.95±0.11	76.28±0.44	84.80±2.09	80.22±0.16	69.84±0.15	0.5389±0.0025
SCAE	52.48±1.67	39.53±0.43	27.95±0.87	52.36±1.23	31.10±0.22	61.98±0.53	11.56±0.26	56.55±0.29	44.60±0.35	0.2289±0.0051
SDAE	22.46±0.02	57.88±0.06	84.27±0.02	69.03±0.37	21.14±0.11	71.67±0.13	46.61±0.52	62.34±0.15	57.36±0.14	0.3733±0.0017
Alexnet	52.95±0.42	**87.54±0.17**	70.05±0.36	81.97±0.22	66.97±0.50	79.87±0.25	94.70±0.16	81.25±0.38	79.40±0.19	0.6740±0.0018
Caffenet	67.54±0.27	82.81±0.15	81.97±0.29	83.13±0.13	70.52±0.38	84.69±0.21	91.54±0.18	76.91±0.32	80.48±0.15	0.6991±0.0012
VGG-S	46.00±0.22	75.30±0.09	78.85±0.14	**93.13±0.36**	79.92±0.13	85.53±0.38	92.64±0.26	64.21±0.42	76.26±0.31	0.6339±0.0019
VGG-16	42.31±0.16	72.65±0.21	81.12±0.14	91.78±0.09	**83.98±0.21**	82.16±0.18	92.33±0.36	62.18±0.14	74.14±0.20	0.6004±0.0017
CNN	**91.83±0.23**	66.70±0.02	75.49±0.11	85.71±0.37	51.56±0.23	**93.44±0.20**	**94.81±0.27**	77.77±0.02	76.79±0.14	0.6635±0.0021
GHCNN	82.94±0.41	80.64±0.35	83.13±0.21	83.23±0.53	64.73±0.93	87.94±0.07	92.28±0.17	**82.43±0.54**	**81.91±0.40**	**0.7267±0.0058**

small-scale SAR image classification task, deeper models appear inefficient and uneconomical. In contrast, the proposed shallow model achieves a cost-effective performance.

Intuitively, the different classification maps of involved algorithms on MDJ data set are shown in Fig. 9. It can be observed that speckle noise brings an enormous impact on SAR image classification. The classification maps of SVM, RF, GBDT, and SDAE algorithms are seriously polluted by the misclassified noisy points. Fuzzy confusion of the non-NN classified maps is mainly distributed in the building area, while also distributed in the farmland area for the SDAE algorithm. However, the deep models have huge advantages in dealing with these problems.

2) Classification Experiments on the DY Data Set: The DY data set covers a medium-size area with regular boundaries of land cover types. In our experiments, ten different algorithms are adopted to compare with the proposed classification framework, including SVM, RF, GBDT, SCAE, SDAE, baseline CNN, Alexnet, Caffenet, VGG-S, and VGG-16. The optimal hyperparameter settings are presented in Table V. All experiments follow the standard specification that multiple redundancy-repeating runs introduce the Monte Carlo random state and data shuffle to ensure the independent trial process.

As shown in Table VII, we note that the overall performance of GHCNN algorithm achieves much superiority to that of others. Specifically speaking, the proposed algorithm has a balanced performance on the pattern recognition of different land cover types. In the building area, shallow models possess better feature-preserving capability than that of deep models. Moreover, it can be noted that the Alexnet and Caffenet models work efficiently as well in the overall classification results. The strategy of greedy layer-wise pretraining has a more dominant promotion on the complex and big data set, while the effectiveness decreases for the simple and small data set. Similar to what we have observed earlier, the non-NN machine learning algorithms have an advantage in water extraction. What calls for special attention is that the unsupervised feature extractor SCAE has a poor performance on SAR image classification. The computational efficiency of involved algorithms is shown in Fig. 12. The depth of different deep models has a great influence on the cost of computational resource. The elapsed time of VGG-16 is nearly seven times than that of the proposed model. It is found that the deeper is not always the better. According to the specific tasks, the optimized selection of deep models is extremely essential.

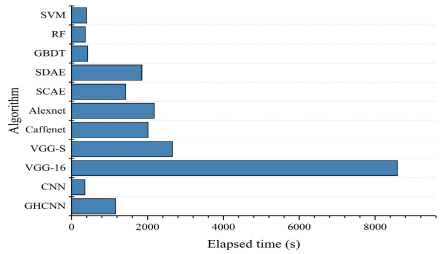


Fig. 12. Total elapsed time of entire classification procedure using different algorithms on region B data set.

As shown in Fig. 11, a series of classification maps is presented intuitively. In the results of non-NN algorithms, commission errors mainly occur between building and water, while omission errors mainly exist in the regions of building and woodland. In terms of baseline CNN, omission errors are reflected in farmland and water areas, and commission errors distribute in building and woodland. As for other deep models, omission errors are mainly reflected in building area, while most of commission errors distribute in woodland area. Obviously, omission errors of GHCNN are at a lower level, and the confusion of building and woodland causes the main commission errors.

3) Classification Experiments on the ZS Data Set: As shown in Table I, the ZS data set covers the minimum region in the three given data sets. Compared with the above two data sets, this data set contains larger areas of water. In the experiments of ZS data set, we adopt ten different contrast algorithms to evaluate the effectiveness and accuracy. All of the algorithms execute the Monte Carlo runs and data shuffle operations to avoid correlation among multiple duplicate experiments. The contrast algorithms and hyperparameter settings are shown in Table V in detail. Moreover, the accuracy comparison, classification maps, and run time are presented in Table VIII and Figs. 13 and 14, respectively.

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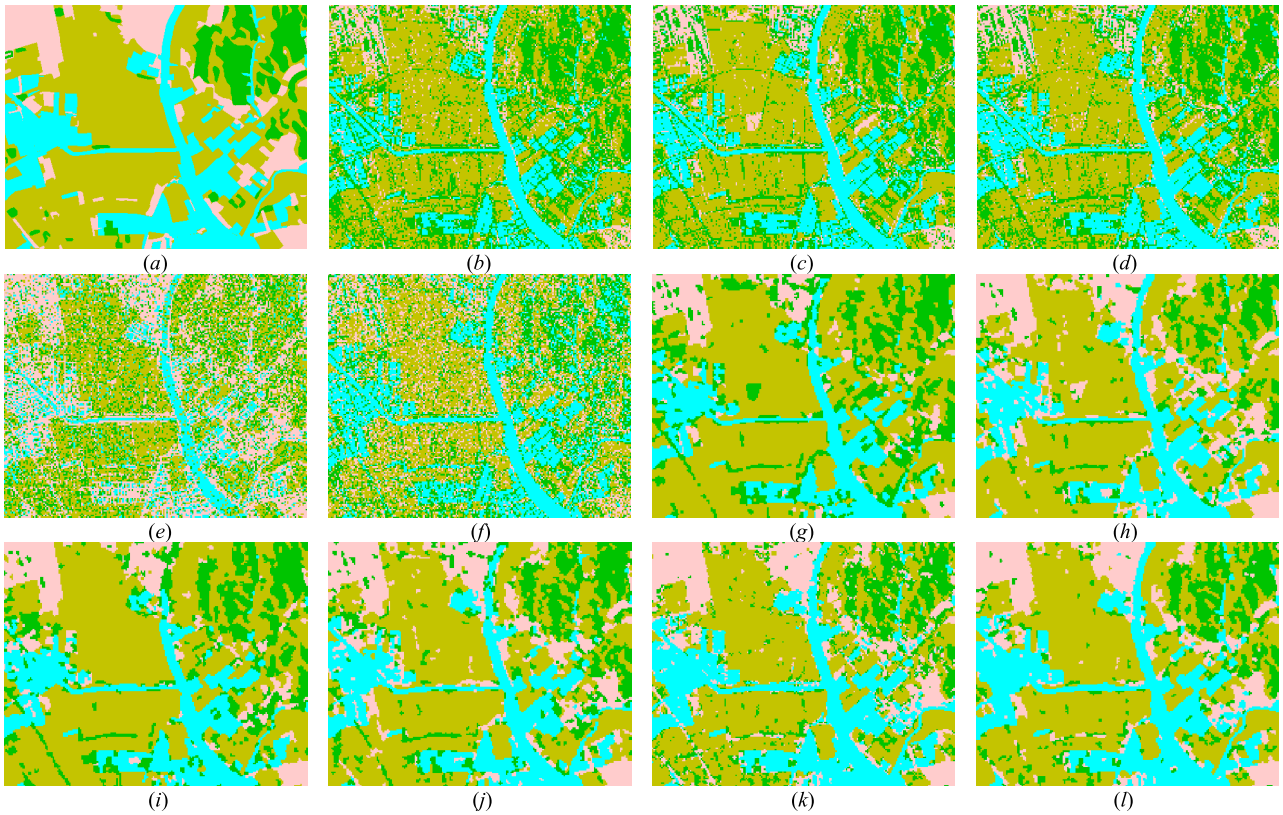


Fig. 13. Classification maps of involved algorithms on the ZS data set. (a) Ground truth. (b) SVM. (c) RF. (d) GBDT. (e) SCAE. (f) SDAE. (g) Alexnet. (h) Caffenet. (i) VGG-S. (j) VGG-16. (k) CNN. (l) GHCNN.

TABLE VIII
ACCURACY COMPARISON OF CLASSIFICATION ON ZS DATA SET

Algorithm	PA				UA				OA	Kappa
	Building	Farmland	Water	Woodland	Building	Farmland	Water	Woodland		
SVM	35.35±0.46	68.61±0.42	70.25±0.51	**77.80±0.16**	62.46±0.94	84.42±0.04	90.08±0.28	25.10±0.11	64.77±0.24	0.4951±0.0026
RF	40.80±0.67	74.55±0.20	77.96±0.22	75.06±0.21	66.97±0.73	88.51±0.06	89.63±0.07	27.96±0.07	70.15±0.05	0.5665±0.0004
GBDT	43.41±0.49	71.68±0.22	80.90±0.07	77.28±0.36	67.74±0.62	89.07±0.21	89.48±0.11	28.11±0.05	69.97±0.03	0.5681±0.0001
SCAE	72.49±2.62	50.68±0.05	36.31±1.30	37.16±2.66	33.27±0.39	79.60±1.55	71.69±3.12	17.26±0.63	49.40±0.34	0.3067±0.0081
SDAE	36.13±0.74	50.58±0.03	74.58±0.40	53.67±1.39	28.94±0.79	82.04±0.71	71.21±1.21	21.83±0.01	54.12±0.36	0.3683±0.0063
Alexnet	54.54±0.38	**95.90±0.19**	73.70±0.28	45.04±0.09	**86.03±0.22**	79.14±0.33	**97.09±0.29**	44.09±0.21	79.27±0.17	0.6615±0.0021
Caffenet	81.48±0.10	84.75±0.29	78.74±0.22	50.99±0.14	64.58±0.37	86.57±0.24	95.32±0.16	46.46±0.37	79.44±0.23	0.6862±0.0036
VGG-S	58.26±0.45	88.71±0.31	71.71±0.18	70.60±0.17	80.05±0.29	85.28±0.13	94.38±0.32	40.43±0.29	78.29±0.31	0.6651±0.0027
VGG-16	80.87±0.28	80.79±0.12	80.34±0.40	69.41±0.09	67.64±0.33	90.55±0.14	93.54±0.08	44.37±0.26	79.54±0.15	0.6961±0.0018
CNN	83.30±0.46	78.24±0.13	91.24±0.23	69.34±0.75	68.44±0.40	**92.89±0.23**	88.95±0.35	49.10±0.26	81.07±0.26	0.7212±0.0040
GHCNN	**85.86±0.64**	84.71±0.24	**91.62±0.24**	74.44±0.55	78.77±0.43	92.50±0.26	90.75±0.41	**58.68±0.39**	**85.41±0.33**	**0.7806±0.0050**

According to Table VIII, it is worth noting that the proposed algorithm is competitive in almost all of the accuracy indexes on the SAR classification task. But there is an exception that the SVM algorithm acquires the best performance on woodland extraction in terms of the PA. The omission errors of non-NN algorithms mainly distribute in building and farmland, while the commission errors are reflected in the building and woodland areas. Compared with the baseline CNN model, the advantages of GHCNN model mainly concentrate on the recognition capability of woodland and farmland areas. The confusion of farmland and woodland contributes to the most of omission and commission errors for the baseline CNN model. Compared with other deep models, the proposed model shows a better performance on woodland and water regions. In Fig. 14, it is clearly described that the deep models need high requirements on time complexity. The proposed shallow model is more cost-effective in the application of SAR image classification.

Visually, it is also obvious in Fig. 13 that the convolutional deep models outperform other models in terms of speckle noise suppression, algorithm robustness, and generalization ability. As for nonconvolution models, the widely distributed

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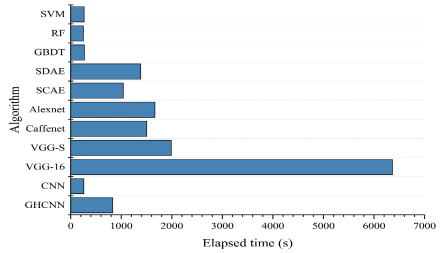


Fig. 14. Total elapsed time of entire classification procedure using different algorithms on region C data set.

SAR geometric distortion introduces the fuzzy confusion to building area, such as the layover and shadow. On the contrary, the convolutional deep models have significant advantages in the local pattern recognition of these complex regions.

IV. Conclusion

High-resolution SAR image classification is always a hot and challenging topic in the fundamental research of microwave remote sensing, especially for the broad SAR image with wide coverage. Inspired by the recent impressive success in data mining on natural images, we design a series of contrast experiments to discuss the adaptation and optimization of deep learning models in the application of SAR image classification. In this article, an original GHCNN algorithm framework is proposed to realize a hierarchical parameter-oriented learning. It can successfully obtain the optimized and effective feature representation, relieve the issues of the confusion derived from speckle noise, and promote the local pattern recognition of geometric distortion in single-polarized SAR image classification. Our semisupervised training pipeline introduces the unsupervised SCAEs as a parameter-oriented preconditioner to the baseline CNN model. This strategy dramatically reduces the empirical nonorientation parameter training from scratch with minimal computational cost.

According to the experimental results and analyses, it is clearly demonstrated that our proposed algorithm framework presents the obvious superiority to the non-NN models. And it also shows competitive advantages over the NN models with full supervision. Moreover, the synchronous efficiency analysis of feature representation using t-SNE algorithm also provides a new perspective for understanding of the black-box nonlinearity during training neural networks. It is worth noting that the greedy hierarchical parameter-oriented models benefit a lot from the unsupervised parameter space, and tend to achieve fast addressing to the global minima for SAR pattern recognition.

In order to advance SAR intelligent interpretation for capturing the ever-changing geoinformation, the follow-up research may consider the introduction of polarimetric SAR data sets, which provide multiple scattering matrices by means of transferring polarized T/R channels.

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References

[1] Y. Bai *et al.*, "A framework of rapid regional tsunami damage recognition from post-event TerraSAR-X imagery using deep neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 1, pp. 43–47, Jan. 2018.

[2] S. Singha, M. Johansson, N. Hughes, S. M. Hvidegaard, and H. Skourup, "Arctic sea ice characterization using spaceborne fully polarimetric L-, C-, and X-band SAR with validation by airborne measurements," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 3715–3734, Jul. 2018.

[3] T. Zhu, F. Li, G. Heygster, and S. Zhang, "Antarctic sea-ice classification based on conditional random fields from RADARSAT-2 dual-polarization satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 6, pp. 2451–2467, Jun. 2016.

[4] J. Li, C. Wang, S. Wang, H. Zhang, Q. Fu, and Y. Wang, "Gaofen-3 sea ice detection based on deep learning," in *Proc. Prog. Electromagn. Res. Symp.-Fall (PIERS-FALL)*, Singapore, Nov. 2017, pp. 933–939.

[5] F. Del Frate, D. Latini, and V. Scappiti, "On neural networks algorithms for oil spill detection when applied to C- and X-band SAR," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Fort Worth, TX, USA, Jul. 2017, pp. 5249–5251.

[6] G. Suresh, C. Melsheimer, J.-H. Korber, and G. Bohrmann, "Automatic estimation of oil seep locations in synthetic aperture radar images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 8, pp. 4218–4230, Aug. 2015.

[7] C. O. Dumitru, S. Cui, G. Schwarz, and M. Datcu, "Information content of Very-High-Resolution SAR images: Semantics, geospatial context, and ontologies," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 4, pp. 1635–1650, Apr. 2015.

[8] A. Tombak, I. Turkmenli, E. Aptoula, and K. Kayabol, "Pixel-based classification of SAR images using feature attribute profiles," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 4, pp. 564–567, Apr. 2019.

[9] M. Ohki and M. Shimada, "Large-area land use and land cover classification with quad, compact, and dual polarization SAR data by PALSAR-2," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5550–5557, Sep. 2018.

[10] H. Wang, R. Magagi, and K. Goita, "Polarimetric decomposition for monitoring crop growth status," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 6, pp. 870–874, Jun. 2016.

[11] S. Skakun, N. Kussul, A. Y. Shelestov, M. Lavreniuk, and O. Kussul, "Efficiency assessment of multitemporal C-band Radarsat-2 intensity and Landsat-8 surface reflectance satellite imagery for crop classification in ukraine," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 8, pp. 3712–3719, Aug. 2016.

[12] L.-K. Soh and C. Tsatsoulis, "Texture analysis of SAR sea ice imagery using gray level co-occurrence matrices," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 2, pp. 780–795, Mar. 1999.

[13] A. Dasgupta, S. Grimaldi, R. Ramsankaran, and J. P. Walker, "Optimized glcm-based texture features for improved SAR-based flood mapping," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Fort Worth, TX, USA, Jul. 2017, pp. 3258–3261.

[14] D. Dai, W. Yang, and H. Sun, "Multilevel local pattern histogram for SAR image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 2, pp. 225–229, Mar. 2011.

[15] M. Picco and G. Palacio, "Unsupervised classification of SAR images using Markov random fields and $\mathcal{G}_I^0$ model," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 2, pp. 350–353, Mar. 2011.

[16] A. Wang, P. Liu, and C. Xie, "Urban land use classification from high-resolution SAR images based on multi-scale Markov random field," in *Proc. 24th Int. Conf. Geoinformatics*, Galway, Ireland, Aug. 2016, pp. 1–4.

[17] A. Voisin, V. A. Krylov, G. Moser, S. B. Serpico, and J. Zerubia, "Classification of very high resolution SAR images of urban areas using copulas and texture in a hierarchical Markov random field model," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 1, pp. 96–100, Jan. 2013.

[18] X. Su, C. He, Q. Feng, X. Deng, and H. Sun, "A supervised classification method based on conditional random fields with multiscale region connection calculus model for SAR image," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 3, pp. 497–501, May 2011.

[19] K. Wang, X. Li, and J. C. Trinder, "Mathematical morphological analysis of typical cyclone eyes on ocean SAR," in *Proc. IEEE Geosci. Remote Sens. Symp.*, Quebec City, QC, Canada, Jul. 2014, pp. 4393–4396.

[20] B. Huang, B. Zhao, and Y. Song, "Urban land-use mapping using a deep convolutional neural network with high spatial resolution multispectral remote sensing imagery," *Remote Sens. Environ.*, vol. 214, pp. 73–86, Sep. 2018.

[21] Y. Xiang, F. Wang, L. Wan, and H. You, "An advanced multiscale edge detector based on Gabor filters for SAR imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 9, pp. 1522–1526, Sep. 2017.

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[22] J. Deng, Y. Ban, J. Liu, L. Li, X. Niu, and B. Zou, "Hierarchical segmentation of multitemporal RADARSAT-2 SAR data using stationary wavelet transform and algebraic multigrid method," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 7, pp. 4353–4363, Jul. 2014.

[23] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, Carson, CA, USA, vol. 60, no. 2, 2012, pp. 1097–1105.

[24] A. Graves, A.-R. Mohamed, and G. Hinton, "Speech recognition with deep recurrent neural networks," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, Vancouver, BC, Canada, May 2013, pp. 6645–6649.

[25] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, Montreal, QC, Canada, vol. 2014, pp. 2672–2680.

[26] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[27] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, pp. 436–444, May 2015.

[28] S. De, L. Bruzzone, A. Bhattacharya, F. Bovolo, and S. Chaudhuri, "A novel technique based on deep learning and a synthetic target database for classification of urban areas in PolSAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 1, pp. 154–170, Jan. 2018.

[29] J. Geng, J. Fan, H. Wang, X. Ma, B. Li, and F. Chen, "High-resolution SAR image classification via deep convolutional autoencoders," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 11, pp. 2351–2355, Nov. 2015.

[30] N. Wang, Y. Wang, H. Liu, Q. Zuo, and J. He, "Feature-fused SAR target discrimination using multiple convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1695–1699, Oct. 2017.

[31] J. Hu, L. Mou, A. Schmitt, and X. X. Zhu, "FusioNet: A two-stream convolutional neural network for urban scene classification using PolSAR and hyperspectral data," in *Proc. Joint Urban Remote Sens. Event (JURSE)*, Event, Dubai, Mar. 2017, pp. 1–4.

[32] J. Geng, H. Wang, J. Fan, and X. Ma, "Deep supervised and contractive neural network for SAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 4, pp. 2442–2459, Apr. 2017.

[33] Z. Ren, B. Hou, Z. Wen, and L. Jiao, "Patch-sorted deep feature learning for high resolution SAR image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 9, pp. 3113–3126, Sep. 2018.

[34] D. Erhan, Y. Bengio, A. Courville, P.-A. Manzagol, P. Vincent, and S. Bengio, "Why does unsupervised pre-training help deep learning?" *J. Mach. Learn. Res.*, vol. 11, pp. 625–660, Feb. 2010.

[35] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Miami, FL, USA, Jun. 2009, pp. 248–255.

[36] Y. Bengio, P. Lamblin, D. Popovici, and H. Larochelle, "Greedy layer-wise training of deep networks," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, Vancouver, BC, Canada, 2007, pp. 153–160.

[37] L. Arnold, H. Paugam-Moisy, and M. Sebag, "Unsupervised layer-wise model selection in deep neural networks," in *Proc. Conf. Ecai, Eur. Conf. Artif. Intell.*, Lisbon, Portugal, 2010. [Online]. Available: https://dl.acm.org/doi/10.5555/1860967.1861146

[38] G. E. Hinton, "Reducing the dimensionality of data with neural networks," *Science*, vol. 313, no. 5786, pp. 504–507, Jul. 2006.

[39] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, Nov. 1998.

[40] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 770–778.

[41] A. Dosovitskiy, J. T. Springenberg, and T. Brox, "Learning to generate chairs with convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Boston, MA, USA, Jun. 2015, pp. 1538–1546.

[42] G. E. Hinton, S. Osindero, and Y.-W. Teh, "A fast learning algorithm for deep belief nets," *Neural Comput.*, vol. 18, no. 7, pp. 1527–1554, Jul. 2006.

[43] L. Bottou and P. Gallinari, "A framework for the cooperation of learning algorithms," in *Proc. Adv. Neural Inf. Process. Syst.*, 1991, pp. 781–788.

[44] S. Offe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn.*, Lille, France, 2015, pp. 448–456.

[45] W. Sun and F. Su, "Regularization of deep neural networks using a novel companion objective function," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Quebec City, QC, Canada, Sep. 2015, pp. 2865–2869.

[46] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 1929–1958, 2014.

[47] A. Romero, C. Gatta, and G. Camps-Valls, "Unsupervised deep feature extraction for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1349–1362, Mar. 2016.

[48] Y. Wei, H. Xiao, H. Shi, Z. Jie, J. Feng, and T. S. Huang, "Revisiting dilated convolution: A simple approach for weakly- and semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 7268–7277.

[49] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[50] L. van der Maaten and G. Hinton, "Visualizing high-dimensional data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, pp. 2579–2605, Nov. 2008.

[51] C.-C. Chang and C.-J. Lin, "LIBSVM: A library for support vector machines," *ACM Trans. Intell. Syst. Technol.*, vol. 2, no. 3, pp. 1–27, Apr. 2011.

[52] L. Breiman, "Random forests," in *Machine Learning*. Dordrecht, The Netherlands: Kluwer, 2001, sec. 45, pp. 5–32.

[53] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *Ann. Statist.*, vol. 29, no. 5, pp. 1189–1232, Oct. 2001.

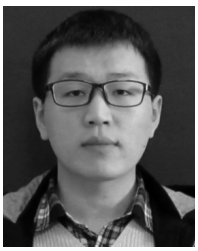
[54] J. Masci, U. Meier, D. Ciresan, and J. Schmidhuber, "Stacked convolutional auto-encoders for hierarchical feature extraction," in *Proc. Artif. Neural Netw. Mach. Learn. 21st Int. Conf. Artif. Neural Netw. (ICANN)*. Espoo, Finland: Springer-Verlag, 2011, pp. 52–59.

[55] P. Vincent, H. Larochelle, Y. Bengio, and P.-A. Manzagol, "Extracting and composing robust features with denoising autoencoders," in *Proc. 25th Int. Conf. Mach. Learn. (ICML)*, 2008, pp. 1096–1103.

[56] Y. Jia *et al.*, "Caffe: Convolutional architecture for fast feature embedding," 2014, *arXiv:1408.5093*. [Online]. Available: http://arxiv.org/abs/1408.5093

[57] K. Chatfield, K. Simonyan, A. Vedaldi, and A. Zisserman, "Return of the devil in the details: Delving deep into convolutional nets," in *Proc. Brit. Mach. Vis. Conf.*, Nottingham, U.K., 2014. [Online]. Available: https://arxiv.org/abs/1405.3531

[58] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*. [Online]. Available: http://arxiv.org/abs/1409.1556



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